

Continuation Under Constraint

Distinctions, Substrates, and Repair Across
Transformations

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Abstract

Placeholder abstract text.

Note on Epistemic Status

This monograph distinguishes, throughout and without exception, between results that are proved, results that are conjectured, questions that are left explicitly open, and claims that are stated informally, ahead of formalization, as a working thesis rather than as any of the above. Five labels are used accordingly. A *Theorem* is a claim with a complete proof given in the chapter or in Appendix A. A *Proposition* is a claim with a complete, if sometimes short, proof, of lesser generality or centrality than a Theorem. A *Conjecture* is a claim believed true, motivated by argument or by worked examples, but not proved in this monograph. An *Open Question* is a question the monograph raises and explicitly declines to resolve, together with a statement of what resolving it would require. A *Principle* is a pre-formal statement of a thesis — most importantly the informal statement of global insufficiency in Chapter 1 and its restatement in Chapter 5 — introduced before the vocabulary needed to state it precisely has been built, and superseded once a formal Proposition or Theorem makes the same claim rigorously; it carries no claim to proof of its own.

Two results carry the label *Theorem*: the Redistribution Theorem (theorem 4.6), imported without modification from *Conservation of Ambiguity* and explicitly scoped there as a synchronic, single-observer result; and the Composition Failure Theorem (theorem 9.2), the new diachronic, multi-agent result

this monograph exists to establish. Every other theorem-like claim in the book is labeled Proposition, Corollary, Conjecture, or Open Question according to the criteria above, and Appendix F consolidates every item labeled Conjecture or Open Question in one place, together with every substantive open item flagged in a Remark rather than a formal environment, so that the book's unproved commitments are never left implicit.

The single largest such commitment is the status of *MEM*| δ 's relationship between memory and representation (chapter 11), which this monograph treats as a genuine, unresolved fork rather than resolving by assumption. Readers should treat Part III's results as conditional on the narrower of the two available readings until that fork is settled by further work.

Relation to Prior Work

This monograph sits downstream of four prior works, from which it imports definitions and one theorem without modification, and against which it states its own extensions precisely.

From *Observability Is Restorability* this monograph imports the basic repair-theoretic vocabulary: repair operators, the repair-stable domain $\text{Rest}(\mathcal{R})$, and the repair interference functional $I(d_i, d_j)$. From *Admissibility Before Truth* it imports the principle that truth (and, in this monograph, sufficiency) is defined only over repair-stable distinctions. From *The Geometry of Reconstruction* it imports the observation that repair operators themselves require higher-order maintenance, motivating this monograph’s treatment of confidence profiles as themselves renderings subject to loss (proposition 8.7).

From *Conservation of Ambiguity* this monograph imports the Redistribution Theorem in full and without modification (theorem 4.6), together with its explicit corollary that ambiguity conservation is only a degenerate limiting case of the more general repair-budget conservation. This monograph does *not* claim to be a corollary of that theorem. The Redistribution Theorem governs a single observer’s competing distinctions at one moment; this monograph’s Composition Failure Theorem (theorem 9.2) governs a single distinction’s status as it moves through a sequence of independently-budgeted stages over time. Appendix C gives the side-by-side comparison in full, including a worked demonstration that one result is not

derivable from the other without the additional budget-relay machinery developed in Part II.

Distinguishability Geometry and Admissibility Theory are imported more informally, as the native vocabulary for, respectively, what survives a projection (chapter 2) and what continuations remain reachable given a substrate (chapter 3). Neither prior work is modified here; both are specialized to the continuation problem this monograph addresses.

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CONTENTS

Part I

Foundations: Distinctions, Substrates, and the Problem of Continuation

Chapter 1

The Continuation Problem

Synopsis. This chapter introduces the question motivating the entire monograph: how can a future consumer continue a task when only a finite artifact remains available? Rather than asking whether a substrate represents its source faithfully in the abstract, the chapter shifts attention to continuation — the practical question of what actions remain possible given a surviving substrate. Chess, painting, and media pipelines are introduced as the recurring examples that generated the framework, before the chapter states its central thesis: no finite substrate can be guaranteed sufficient for arbitrary future continuation across an open task space, because future requirements cannot be fully anticipated at the time the substrate is produced.

1.1 A Motivating Scenario

An artifact is produced. A consumer arrives later, possibly much later, possibly through an intermediary the producer never anticipated, and must act on the basis of what is available to them. The question this monograph asks is simple to state and difficult to resolve: what must survive the interval between production and use for that later action to succeed?

1.2 Informal Statement of the Thesis

Principle 1.1 (Global Insufficiency, informal form). No finite substrate can be guaranteed sufficient for arbitrary future continuation across an open task space.

This statement is deliberately informal at this stage. Chapter 3 gives it a precise form as proposition 3.7, and Part II extends it from a single substrate to a chain of substrates, where the monograph's central new theorem is proved (theorem 9.2).

1.3 Continuation, Not Representation

Continuation is defined operationally throughout this monograph: given only a substrate r , the set of actions a consumer can correctly perform in pursuit of a task τ . This is deliberately narrower than asking whether r is a faithful, meaningful, or complete representation of some original object e in the abstract. Faithfulness-to-an-original is not directly measurable by a consumer who never had access to e ; sufficiency-for-a-task is.

1.4 Three Motivating Domains

Three domains recur throughout the monograph and are introduced here only informally, to be formalized in Part IV. A chess

position illustrates a domain with closed, enumerable hidden state: what is missing from the visible board (castling rights, en passant eligibility, the repetition count) can in principle be listed exhaustively and restored. A painting illustrates the opposite case: a static configuration, such as a branch bent as if by wind, is compatible with an open-ended and non-enumerable set of causal histories, and no finite augmentation closes that space. A media pipeline – audio rendered into a transcript, a transcript compressed into a summary – illustrates a domain in which loss accumulates across a sequence of independently produced renderings, each locally reasonable, with no shared record of what any earlier stage discarded. These three domains are not illustrations selected after the fact; they are the cases that generated the framework, and each returns with its own full chapter in Part IV.

1.5 Roadmap

Part I develops the vocabulary needed to state principle 1.1 precisely: distinguishability loss (chapter 2), admissible continuation (chapter 3), and repair economics (chapter 4), before Chapter 5 identifies exactly where the imported machinery stops applying. Part II introduces the budget relay, the monograph's new formal object, and proves the Composition Failure Theorem. Part III resolves the preservation/reconstitution question raised by the informal motivating conversation behind this project, correcting its initial binary framing into a continuum. Part IV returns to chess, painting, and the rest with the full apparatus available. Part V closes with a single statement tying distinguishability, admissibility, repair budgets, and continuation together, and an explicit accounting of what remains unresolved.

Chapter 2

Distinctions and Distinguishability Loss

Synopsis. This chapter establishes the formal vocabulary used throughout the book. Drawing on Distinguishability Geometry, it treats information as a structured space of distinctions and describes renderings as projections that preserve some distinctions while collapsing others. The chapter’s central move is the distinction between benign underdetermination and corruption: not all loss is harmful, and corruption arises not from loss alone but from *undeclared* loss that obscures what has been discarded. This reframes information quality around disclosure rather than preservation alone.

2.1 Distinguishability Geometry, Imported

Distinguishability Geometry supplies this monograph’s most basic object: a space of distinctions $(X, \sim_X)$, together with an equivalence structure recording what is and is not told apart

within it. Nothing in this section modifies that prior formalism; it is specialized here to the case where X is the distinction-space of an object e under transformation.

Remark 2.1 (A Standing Simplification). Throughout this monograph, distinction spaces are treated as finite or countable, and the cardinality-based measures used later (definition 10.2 in particular) are stated only for that case. This is a simplification made for tractability, not a claim that every domain of interest is discrete in the relevant sense; a painting’s causal-history space, for instance, is arguably continuous rather than merely large. Where this matters, it is flagged again locally (chapter 14); it is not resolved here.

2.2 Renderings as Projections

Definition 2.2 (Rendering as Projection). A rendering r of an object e is a map $(X, \sim_X) \rightarrow (Y, \sim_Y)$ under which some distinctions separated in X become identified in Y .

This definition is stated at the level of Distinguishability Geometry’s own primitives: X and Y are sets of possible states, and $\sim_X, \sim_Y$ record which pairs of states are, and are not, told apart within each. This is the correct level of generality for the formalism, but it is not the level at which the rest of this monograph actually talks. From Chapter 3 onward, and especially throughout Chapter 6, this book speaks of *a* distinction d – “distinction d_3 ,” “the repair cost of d ,” “ d is dropped at stage i ” – as though a distinction were a single named thing, not a pair of states. Reconciling these two ways of talking, rather than quietly switching between them, is this chapter’s principal piece of technical work.

2.2.1 From Pairs to Named Distinctions

Definition 2.3 (Discarded Pair). Given a rendering $r : (X, \sim_X) \rightarrow (Y, \sim_Y)$ realized by an underlying map $\varphi : X \rightarrow Y$, an unordered pair $\{x, x'\} \subseteq X$ is a *discarded pair* of r if $x \not\sim_X x'$ (the pair is told apart in the source) but $\varphi(x) \sim_Y \varphi(x')$ (the pair becomes indistinguishable under r).

definition 2.3 is the pair-level primitive Distinguishability Geometry actually provides, and it is the correct formal object underlying everything that follows. It is, however, an awkward unit to reason about directly: nobody asks whether “the pair {recording with emphasis on the third syllable, recording without it}” was preserved; they ask whether *emphasis* was preserved. The bridge from pairs to named features is given by treating a feature as the thing that distinguishes a minimal pair.

Definition 2.4 (Minimal Pair for a Distinction). A named distinction d is represented, relative to a source $(X, \sim_X)$, by a minimal pair $\{x_d, x_{\neg d}\} \subseteq X$: two states differing exactly in whether the feature named by d holds, and told apart in the source ($x_d \not\sim_X x_{\neg d}$).

Definition 2.5 (Discarded Distinction Set, Revised). Given a rendering r and a fixed assignment of minimal pairs to named distinctions (definition 2.4), the discarded distinction set $D(r)$ is the set of named distinctions d whose minimal pair $\{x_d, x_{\neg d}\}$ is a discarded pair of r (definition 2.3):

$$D(r) = \{d : \{x_d, x_{\neg d}\} \text{ is discarded by } r\}.$$

definition 2.5 supersedes an earlier, looser formulation of the same definition, which described $D(r)$ directly as “the set of distinction-pairs separated in the source but identified in r ” and was consequently ambiguous between the pair-level and feature-level readings used elsewhere in the book. Every

later chapter’s usage – definition 2.7 and definition 2.8 below, Chapter 4’s treatment of $C_R(d)$ for a named d , and the entire apparatus of Chapter 6 – presumes the feature-level reading given here, and definition 2.5 is stated so as to make that presumption explicit and correct rather than convenient and unexamined.

Remark 2.6 (When the Bridge Is Clean). The identification in definition 2.5 between a discarded pair and a discarded feature is unproblematic exactly when X factors, at least locally, as a product of independent feature-dimensions – so that varying d alone, holding everything else fixed, is a meaningful operation on X and produces a genuine minimal pair rather than an artifact of an arbitrary choice of coordinates. This holds cleanly for the toy and worked examples used throughout this monograph (a transcript’s lexical content and prosodic emphasis are reasonably treated as independent dimensions of an audio recording) and is assumed as a standing simplification rather than argued for in general. Domains where features are not independently variable in this sense – where, for instance, no change to one property is possible without simultaneously changing another – are not treated here.

2.3 Underdetermination Versus Corruption

The distinction developed in this section is used, in one form or another, in every subsequent chapter of the monograph, and is the closest thing this book has to a single organizing move.

Definition 2.7 (Benign Loss). A discarded distinction $d \in D(r)$ is benign relative to a task τ if d is not required to perform τ correctly.

Definition 2.8 (Corrupting Loss). A discarded distinction $d \in D(r)$ is corrupting relative to τ if τ requires d and the rendering’s presentation obscures that d was discarded, as opposed to openly declaring the gap.

Proposition 2.9 (Corruption Requires Undeclared Loss). *Let r be a rendering with discard set $D(r)$. If every distinction in $D(r)$ is declared – that is, made available to the consumer as a known gap rather than presented as preserved – then loss alone does not imply corruption. Corruption requires both (i) the loss of a task-relevant distinction, and (ii) failure to disclose that loss.*

Proof Sketch. Corruption, per definition 2.8, is task-relative by construction. If $d \in D(r)$ is declared, a consumer pursuing a task requiring d can compensate for its absence, decline to rely on r for that task, or seek an alternative rendering; the distinction remains unavailable, but the consumer’s belief-state about r ’s sufficiency is not falsified by r itself. Undeclared loss is therefore necessary for corruption in the sense of definition 2.8, though not sufficient, since a task not requiring d is unaffected regardless of disclosure.

definition 2.7 and definition 2.8 are stated here in terms of “required to perform τ correctly” without yet saying precisely what correctness of performance means, since that requires the apparatus of admissible continuation sets developed in Chapter 3. proposition 3.6 there closes the loop, restating both definitions in the sharper, task-relative vocabulary Chapter 3 provides.

2.4 A Toy Example

Example 2.10. Lossy image compression discards high-frequency detail below a quantization threshold. When the compression ratio and quantization parameters are attached to the file as metadata, the loss is declared: a consumer needing pixel-exact reproduction can detect, before relying on the image, that it will not get it. When the same file is relabeled as a lossless master, an identical discard set becomes corrupting under definition 2.8, though nothing about the image data itself has

changed. In the vocabulary of definition 2.4, each discarded high-frequency component corresponds to a minimal pair of source images differing only in that component, rendered indistinguishable once quantized; the metadata either does or does not disclose that this minimal pair has been collapsed. The example is deliberately kept toy at this stage; it is developed properly, without simplification, in the media-pipeline case of Chapter 20.

Chapter 3

Admissibility and the Space of Continuations

Synopsis. Having defined what can be lost, this chapter turns to what remains possible. Using concepts imported from Admissibility Theory, it defines admissible continuation sets and task-relative sufficiency: a substrate is never sufficient in the abstract, only relative to a particular task. The chapter argues that sufficiency cannot generally be certified at creation time, since future tasks are often unknown then, and introduces the distinction between closed hidden-state domains, where omitted information can in principle be enumerated and restored, and open domains, where no finite augmentation can exhaust future informational needs. This distinction becomes one of the framework's central organizing principles.

3.1 Admissibility Theory, Imported and Specialized

Admissibility Theory supplies the second basic object this monograph needs: given a substrate, the set of continuations still reachable from it. As with Distinguishability Geometry in Chapter 2, nothing in the prior formalism is altered here; it is specialized to the continuation problem of chapter 1.

Definition 3.1 (Admissible Continuation Set). $\mathcal{A}_\tau(r)$, the admissible continuation set of task τ given rendering r , is the set of continuations of τ that remain reachable given only r .

This definition leaves the internal structure of a “continuation” deliberately unspecified, and deliberately so: a continuation is a legal move in Chapter 13’s setting, a plausible causal narrative in Chapter 14’s, a verified-or-rejected verdict in Chapter 16’s, and a binding ruling in Chapter 17’s. The formalism is domain-agnostic on purpose; each worked domain in Part IV supplies its own concrete continuation space rather than inheriting one from this chapter.

Proposition 3.2 (Admissibility Bounded by Full Access). *For any rendering r of an object e and any task τ , $\mathcal{A}_\tau(r) \subseteq \mathcal{A}_\tau(e)$.*

Proof Sketch. A rendering (definition 2.2) can only identify distinctions present in the source, never introduce new ones; any continuation reachable using only r is therefore reachable using the full source e as well, since e determines everything r does and possibly more. The reverse containment need not hold, which is exactly what makes discard consequential.

Proposition 3.3 (Admissibility Is Monotone Under Further Rendering). *If r' is itself a rendering of r – that is, r' is obtained from r by a further projection in the sense of definition 2.2 – then $\mathcal{A}_\tau(r') \subseteq \mathcal{A}_\tau(r)$.*

Proof Sketch. Immediate from proposition 3.2 applied with r playing the role of the source and r' the rendering of it: r' can only identify distinctions already present or absent in r , and cannot recover any continuation r itself could not reach.

proposition 3.3 is the admissibility-theoretic counterpart of Chapter 6’s proposition 6.12: continuation possibilities can only shrink or stay level as a substrate is further rendered, never expand, in the absence of a preserved earlier stage to re-derive from (proposition 6.9).

3.2 What a Task Requires

Chapter 2 spoke of a distinction being “required” by a task τ without saying precisely what that means. The remainder of this chapter makes that precise, and in doing so corrects an oversimplification present in an earlier formulation of task-relative sufficiency: that formulation spoke of “the true or intended continuation” of a task, in the singular, which is adequate for tasks like verification that admit exactly one correct outcome but is not adequate for tasks like Chapter 14’s evocative reconstruction, which by design tolerates more than one acceptable completion.

Definition 3.4 (Correctness Set). A task τ , evaluated against an object e , determines a nonempty correctness set $\mathcal{C}_\tau(e)$: the set of continuations that count as successfully performing τ . $\mathcal{C}_\tau(e)$ is a singleton for tasks demanding a unique correct outcome (verification against a fixed proof, say) and may contain many continuations for tasks tolerating multiple acceptable completions (evocative reconstruction from a painting, say, where several causal histories are equally acceptable).

Definition 3.5 (Task-Relative Sufficiency, Revised). r is sufficient for τ (evaluated against the true object e) iff

$$\mathcal{A}_\tau(r) \subseteq \mathcal{C}_\tau(e).$$

This revises, and is intended to supersede, the earlier singular-outcome phrasing. The subset formulation in definition 3.5 is chosen because it correctly unifies the two cases an adequate definition needs to cover simultaneously. For a verification-type task, $\mathcal{C}_\tau(e)$ is a singleton, and definition 3.5 demands that r pin down exactly that outcome: $\mathcal{A}_\tau(r)$ must not merely intersect the correct answer but be contained in it, ruling out any admissible but wrong continuation being reachable alongside the right one. For a task tolerating multiple acceptable completions, $\mathcal{C}_\tau(e)$ is larger, and definition 3.5 only requires that every continuation r makes reachable falls inside that larger acceptable set – which is satisfied even when several distinct, mutually incompatible continuations remain open, exactly the situation Chapter 14 calls benign underdetermination.

Proposition 3.6 (Benign Loss Preserves Sufficiency). *If every distinction in $D(r)$ is benign relative to τ (definition 2.7), and e itself would be sufficient for τ , then r is sufficient for τ .*

Proof Sketch. By definition 2.7, no distinction required to determine membership in $\mathcal{C}_\tau(e)$ is among those r discards; hence every continuation in $\mathcal{A}_\tau(r)$ is also determinable from e as belonging to $\mathcal{C}_\tau(e)$, since the discarded distinctions play no role in that determination. $\mathcal{A}_\tau(r) \subseteq \mathcal{A}_\tau(e) \cap (\text{continuations consistent with } \tau) \subseteq \mathcal{C}_\tau(e)$ whenever e is itself sufficient, giving definition 3.5 directly.

proposition 3.6 is the formal statement of the coherence this monograph’s two central chapters are supposed to have with each other: Chapter 2’s loss-level vocabulary (benign versus corrupting) and this chapter’s continuation-level vocabulary (admissible versus sufficient) are not two independent frameworks that happen to agree on examples; benign loss is, by construction, exactly the kind of loss that cannot violate sufficiency.

3.3 Why Sufficiency Cannot Be Verified at Production Time

Proposition 3.7 (Task Unavailability at Ingestion). *If the future task space T relevant to a substrate is open – unbounded, not fixed at the time the substrate is produced – then sufficiency of a rendering r relative to T cannot be certified when r is produced.*

Proof Sketch. Suppose otherwise: suppose r could be certified sufficient for every $\tau \in T$ at production time. Then $\mathcal{A}_\tau(r) \subseteq \mathcal{C}_\tau(e)$ for every $\tau \in T$, which requires r to have preserved every distinction any such τ might need to rule out an incorrect continuation. Since T is open, a task τ^* can always be constructed, after the fact, requiring a distinction not represented in r – for instance, a distinction r 's producer had no reason to anticipate – for which $\mathcal{A}_{\tau^*}(r) \not\subseteq \mathcal{C}_{\tau^*}(e)$. This contradicts the certification assumed above. Hence no such certification is available in general.

This result is the formal version of principle 1.1 for a single substrate. It motivates the shift, completed in definition 6.8, from optimizing a rendering for sufficiency – which proposition 3.7 shows to be generally impossible to verify – to optimizing it for re-derivability from a preserved root object.

3.4 Closed and Open Hidden-State Domains

Definition 3.8 (Closed versus Open Hidden-State Domain). A domain has *closed* hidden state if the distinctions omitted from a given state are drawn from a finite, enumerable list – for instance, a chess position's castling rights, en passant eligibility, and repetition count. A domain is *open* otherwise – for instance, the causal history of a depicted configuration in a painting, which is compatible with an unbounded and non-enumerable set of candidate causes.

Proposition 3.9 (Closed Hidden-State Sufficiency). *In a domain with a finite hidden-state set H , the augmented rendering $r' = (r, H)$ is sufficient for any task τ whose correctness set $\mathcal{C}_\tau(e)$ is determined by distinctions drawn from H , provided all task-relevant elements of H are included.*

Proof Sketch. Since H is finite, exhaustive enumeration eliminates ambiguity about which hidden variables are missing: every continuation's membership in $\mathcal{C}_\tau(e)$ becomes computable from r' once the enumerated variables are fixed, so $\mathcal{A}_\tau(r') \subseteq \mathcal{C}_\tau(e)$ as required by definition 3.5. Chess serves as a witness (Chapter 13): a board position augmented with castling rights, en passant status, and repetition count is sufficient for determining every legal continuation of the game, whereas the visible board alone is not.

Proposition 3.10 (Sufficiency Achievable Only in the Closed Case). *Sufficiency in the sense of proposition 3.9 – achieved by finite augmentation of the hidden state – is available only in closed hidden-state domains.*

Proof Sketch. In an open hidden-state domain (definition 3.8), no finite set H exhausts the candidate causes or hidden variables relevant to a given task, by definition of openness; hence no finite H makes proposition 3.9's hypothesis satisfiable. Painting (Chapter 14) is the canonical witness to this proposition's converse case, and there $\mathcal{C}_\tau(e)$ is typically large rather than singleton, which is exactly what allows sufficiency to be achieved by other means (proposition 3.6) even though augmentation specifically is unavailable.

Chapter 4

Repair, Cost, and the Economics of Maintaining a Distinction

Synopsis. This chapter imports the repair-theoretic machinery of *Conservation of Ambiguity*: repair cost, ambiguity, and interference as quantitative descriptions of how finite budgets constrain distinguishability. The Redistribution Theorem is restated and carefully scoped as a description of how a single observer reallocates finite resources across distinctions within one repair economy at one moment. The chapter's key emphasis, on which the rest of the monograph depends, is that ambiguity conservation is not that theorem's central result; the deeper conserved quantity is the repair budget itself, and ambiguity conservation is recovered only as a restrictive limiting case.

4.1 Repair Cost as the Cost of Preserving a Minimal Pair

Chapter 2 defined a named distinction d via a minimal pair $\{x_d, x_{-d}\}$ (definition 2.4) and defined d as discarded by a rendering precisely when that pair collapses (definition 2.5). This chapter’s imported apparatus speaks instead of the *repair cost* of d , and it is worth stating plainly, before using the two vocabularies interchangeably as every later chapter does, exactly how they connect.

Definition 4.1 (Repair Cost, Specialized). For a named distinction d with minimal pair $\{x_d, x_{-d}\}$ subject to a perturbation process (a transformation, a transmission channel, a compression step), the repair cost $C_R(d)$ is the cost of maintaining $x_d \not\sim x_{-d}$ – that is, of keeping the pair told apart – under that perturbation.

definition 4.1 is a specialization of Chapter 4’s imported notion of repair cost to the minimal-pair vocabulary of Chapter 2, and it is what licenses Chapter 6’s otherwise unexplained move of treating $C_R(d)$ and $D(r)$ as talking about the same underlying phenomenon: a distinction is discarded by a rendering exactly when insufficient repair investment was made, relative to that rendering’s perturbation, to keep its minimal pair apart.

4.2 Repair Cost, Sharpness, and Ambiguity

(Imported from Conservation of Ambiguity.)

Definition 4.2 (Distinction Sharpness). For $d \in \text{Rest}(\mathcal{R})$ a repair-stable distinction, the sharpness of d is $\sigma(d) = C_R(d)$, the repair cost required to maintain d under admissible perturbation.

Definition 4.3 (Operational Ambiguity). The ambiguity of a distinction is a monotone decreasing function of its repair cost; the simplest choice is $A(d) = 1/C_R(d)$.

Remark 4.4 (Ambiguity’s Functional Form Is a Convention, Not a Law). *Conservation of Ambiguity* states this explicitly, and it is worth repeating rather than letting the simplest choice harden into an assumption: any strictly decreasing monotone transformation of $C_R(d)$ produces an equivalent *ordering* of distinctions by ambiguity, so $A(d) = 1/C_R(d)$ is adopted here for concreteness and ease of computation in worked examples, not because it is uniquely forced by anything proved. Nothing in this monograph depends on the specific functional form; every result stated in terms of $A(d)$ could be restated, without loss, in terms of the ordering it induces.

Ambiguity, under definition 4.3, is not ignorance, incomplete information, or subjective uncertainty. It measures the degree to which a distinction has been left under-maintained: a distinction is ambiguous because insufficient repair capacity has been committed to stabilizing it against perturbation, not because information about it is unknowable in principle.

4.3 Repair Interference

Definition 4.5 (Repair Interference). For $d_i, d_j \in \text{Rest}(\mathcal{R})$, the repair interference $I(d_i, d_j)$ is the change in repair cost of d_j induced by an infinitesimal increase in repair investment devoted to d_i . Positive interference indicates repair competition; negative interference indicates repair synergy; zero interference indicates operational independence.

4.4 The Redistribution Theorem

Theorem 4.6 (Redistribution Theorem). (*Imported from Conservation of Ambiguity.*) Let $\Phi = A(\cdot) + \lambda C_R(\cdot) + \mu I(\cdot, \cdot)$ be the repair budget functional of a finite observer, with total budget Φ_{tot} held fixed. Then every admissible variation satisfying $\delta C_R(d_1) > 0$ for some distinction d_1 induces compensating variations throughout the remaining distinction space such that $\delta\Phi = 0$. Consequently, no local increase in distinction sharpness can occur without global redistribution of ambiguity, repair cost, or repair interference.

Proof Sketch. (*Imported from Conservation of Ambiguity.*) The result follows from constrained variation of the repair budget functional Φ under fixed total budget. Since the observer possesses no additional maintenance capacity, every increase in repair investment devoted to one region of distinction space must appear elsewhere as a corresponding change in ambiguity, repair cost, or repair interference. The full proof, including the three interference regimes (competition, synergy, independence), is given in *Conservation of Ambiguity* and is not repeated here.

Scope note. theorem 4.6 governs a single observer's competing distinctions at a single moment: a *synchronic* result. It says nothing, on its own, about what happens to a single distinction's status as it is carried across a sequence of independently-budgeted stages over time. That extension is the subject of Chapter 5 and Part II.

Corollary 4.7 (Ambiguity Conservation as a Limiting Case). (*Imported from Conservation of Ambiguity.*) If repair costs and interference remain fixed ($\delta C_R(\cdot) = \delta I(\cdot, \cdot) = 0$), then $\delta A(\cdot) = 0$: ambiguity conservation is recovered as a special case of repair-budget conservation. In realistic systems, sharpening one distinction almost always changes both repair costs and interference relationships, so ambiguity alone is generally not conserved; the conserved quantity is the complete repair budget Φ .

This corollary is the reason this monograph does not treat itself as a corollary of theorem 4.6: the earlier framing of the research program that produced this book – “ambiguity is conserved, merely relocated” – is precisely the special case corollary 4.7 shows to be exceptionally restrictive and generally false. The new work required here (Part II) does not inherit its central claim from this theorem; it must be argued independently.

4.5 A Worked Instance, Before It Becomes a Relay

Chapter 6 introduces a running numerical example: five distinctions $d_1, \dots, d_5$ with repair costs $C_R(d_1) = 1$, $C_R(d_2) = 2$, $C_R(d_3) = 3$, $C_R(d_4) = 2$, $C_R(d_5) = 4$. There, the costs are held fixed and a *budget* β_i varies from stage to stage of a relay. Here, before any relay exists, it is worth seeing what theorem 4.6 says about the same five distinctions considered as the simultaneous concern of a *single* observer with one fixed total budget Φ_{tot} – the case the theorem actually governs.

Suppose the observer initially funds all five distinctions exactly at their stated costs, for a total commitment of $1 + 2 + 3 + 2 + 4 = 12$, and suppose $\Phi_{\text{tot}} = 12$ exactly, leaving no slack. Now suppose the observer wishes to sharpen d_3 – to invest further in it beyond its baseline cost, driving $C_R(d_3)$ up and $A(d_3)$ down, perhaps because d_3 has become newly important. theorem 4.6 states that this cannot happen for free: with Φ_{tot} fixed at 12 and already fully committed, $\delta C_R(d_3) > 0$ forces a compensating $\delta \Phi = 0$ elsewhere in the functional. If interference is negligible ($I \approx 0$, as in the worked relay of section 6.8), the compensation must come directly out of the repair cost or ambiguity terms of the remaining four distinctions – concretely, some combination of d_1, d_2, d_4, d_5 must have its own funding reduced by exactly the amount d_3 ’s sharpening consumed, raising their ambiguity

in turn.

This is the single-observer, single-moment phenomenon theorem 4.6 describes, and it is worth contrasting directly with what happens to the *same five numbers* once Chapter 6 turns them into a two-stage relay with $\beta_0 = 4$ and $\beta_1 = 3$. There, d_3 's low priority at stage 0 does not trigger any compensating adjustment to d_1 or d_2 's funding elsewhere in a shared Φ_{tot} , because no such shared quantity exists across stages (definition 6.6): d_3 , d_4 , and d_5 are simply dropped, permanently, with nothing anywhere in the system compensating for their loss. The redistribution theorem 4.6 guarantees within one economy has no counterpart at all across the boundary between β_0 and β_1 . Chapter 5 argues for this contrast in prose; proposition 5.1 there states it as a precise claim about why no functional analogous to Φ can be defined across a relay's stages in the first place.

Chapter 5

From Synchronic Budgets to Diachronic Chains

Synopsis. This chapter identifies the conceptual gap that motivates the monograph's primary contribution. Existing repair-budget theory describes trade-offs among distinctions maintained by one observer at one moment; real information systems instead consist of chains of transformations performed by many independent agents across time. Local repair-budget reasoning cannot simply be applied stage by stage, because no global coordination mechanism exists and each stage acts with incomplete knowledge of future tasks and other stages' decisions. This motivates the new formal object introduced in Part II.

5.1 The Mismatch, Stated Precisely

theorem 4.6 governs competition among distinctions maintained by one observer at one time. The continuation problem of Chapter 1 concerns a single distinction, or a bounded clus-

ter of distinctions, whose status must survive transmission across a sequence of observers or processes over time. These are different objects, and the difference is not cosmetic.

5.2 Why This Is Not a Trivial Relabeling

Three features of the continuation problem have no counterpart in the Redistribution Theorem’s setting. First, no single global budget functional constrains an entire chain of independently operating stages; each stage’s budget is its own. Second, each stage’s repair decisions are typically made without visibility into any other stage’s decisions, past or future. Third, the chain is directional: a later stage cannot un-discard a distinction an earlier stage has already collapsed, whereas theorem 4.6’s variational argument implicitly allows reallocation in any direction within a single observer’s simultaneous budget.

The temptation to treat these as differences of degree rather than of kind is worth resisting directly, since a natural first response to the continuation problem is to try applying theorem 4.6 stage-by-stage – to ask what a “local Redistribution Theorem” at each B_i would say, and to hope that composing n local instances yields something like a global conservation law across the whole chain. The remainder of this section shows why that hope fails, and why it fails for a structural reason rather than merely a technical one.

Proposition 5.1 (No Global Conserved Quantity Across a Relay). *There is, in general, no functional $\Phi_{\text{relay}} = f(\beta_0, \dots, \beta_{n-1})$ of a budget relay’s stage budgets satisfying an analogue of theorem 4.6’s conservation law under variation of any single β_i .*

Proof Sketch. theorem 4.6’s conservation law depends essentially on the interference functional $I(\cdot, \cdot)$ being defined over pairs of distinctions competing within a *single* repair-stable

domain $\text{Rest}(\mathcal{R})$, so that a variation in the repair investment devoted to one distinction has a well-defined, computable effect on the repair cost of another via $I(\cdot, \cdot)$ (definition 4.5). Under definition 6.6, however, stage i 's interference function I_i is defined only over distinctions present at stage i ; no interference term $I(d, d')$ is defined, or is even well-typed, for d a distinction competing for β_i and d' a distinction competing for β_j at a different stage j , since the two distinctions need not both survive to be simultaneously present at any single stage for such a term to be evaluated. A variation $\delta\beta_i$ therefore has no defined effect on β_j for $j \neq i$ within this formalism, not merely an uncoordinated or unobserved one: the mathematical object *theorem 4.6's* proof varies – a single Φ built from terms sharing one common domain $\text{Rest}(\mathcal{R})$ – has no counterpart spanning stages, since $\text{Rest}(\mathcal{R})$ itself is stage-local by definition 6.1. No conservation law can be stated for a quantity that has not been defined, and Φ_{relay} has not been defined for exactly this reason.

proposition 5.1 sharpens the informal claim made in earlier drafts of this argument – that “no global coordination mechanism exists” – into a specific mathematical reason rather than a description of a missing institutional arrangement. It is not merely that no one has coordinated the stages of a typical relay; it is that the formal object a coordination claim would need to be about does not exist under definition 6.6's stage-local interference structure. This is also why Chapter 7's partial-coordination mechanisms cannot simply consist of “computing Φ_{relay} and applying theorem 4.6 to it”: that theorem is unavailable in this setting not because the chain is uncoordinated but because the relevant conserved quantity has no definition to conserve. Coordination mechanisms must therefore do something other than extend Chapter 4's apparatus outward; they must introduce new machinery, which is exactly what Chapter 7's taxonomy does.

5.3 Statement of the Extension Task

The remainder of this monograph is organized around building a formal object capable of expressing “locally rational, globally uncoordinated” loss accumulation – the budget relay of Chapter 6 – and proving a composition result analogous in spirit to theorem 4.6 but diachronic in structure: the Composition Failure Theorem of Chapter 9. Given proposition 5.1, that theorem cannot be obtained by generalizing theorem 4.6’s variational argument to a larger domain; it must be proved directly against definition 6.1 and definition 6.2’s irreversibility, as it is in Appendix A.

Principle 5.2 (Global Insufficiency, restated). A substrate chain in which each transformation is locally budget-rational can nonetheless become globally insufficient for an open task space, without any single stage being negligent.

This sentence is, in an important sense, the actual center of the monograph – more so than any single theorem statement that follows from it. It is stated here informally, and precisely, because everything in Part II exists to make it rigorous, and proposition 5.1 is the first indication that making it rigorous will require genuinely new machinery rather than a larger application of what Chapter 4 already provides.

Part II

The Budget Relay: A New Formal Object

Chapter 6

Defining the Budget Relay

Synopsis. This chapter introduces the budget relay, the monograph's primary new theoretical object: a chain of renderings connected by independent local repair economies, each governed by its own costs, priorities, and constraints. The chapter formalizes how distinctions are discarded under local budget pressure and why such losses become irreversible once no record of them remains. Root objects and budget-aware loss ledgers are introduced as the mechanisms for measuring, and in some cases recovering from, downstream insufficiency; preserving a root object is shown to be fundamentally different from preserving every downstream rendering, since it alone permits re-derivation under newly discovered tasks.

This chapter carries disproportionate weight in the monograph relative to its length. The object defined in definition 6.6 is the only genuinely new mathematical object this book introduces; every later theorem is, in an important sense, a property of this definition rather than an independent contribution. If this definition is not judged inevitable and useful on its own

terms, no proof in Chapter 9 will rescue it. The chapter is therefore written at a slower pace than its material would otherwise require, and it deliberately stress-tests the definition against a fully worked numerical instance before the rest of the monograph is permitted to build on it. Two genuine gaps surface in the course of this stress-testing — the underdetermination of the local discard rule, and the bootstrapping cost of preserving a root object — and both are carried forward explicitly into Appendix F rather than quietly absorbed into the definition.

6.1 Why a Chain, and Not Just a Rendering

Chapter 2 defined a rendering as a single projection from a source’s distinction space to a smaller one, and Chapter 3 asked, of a single such rendering, whether it is sufficient for a given task. Neither chapter had reason to ask what happens when a rendering is itself produced from an earlier rendering, and that one from a still earlier one, with a different party responsible for each step and no party responsible for the sequence as a whole. This is not a generalization added for its own sake. It is the ordinary condition of almost every substrate a real consumer actually encounters: a transcript is produced from a recording, not from the recorded event itself; a textbook’s account of a proof is produced from the proof, not independently derived; a legal memorandum summarizing a contract is produced from the contract, which was itself drafted from negotiation notes that no longer exist. Treating each of these as an isolated rendering of some single, fixed original – as Chapters 2 and 3 implicitly do when they speak of “the” discard set of “a” rendering r of “an” object e – silently assumes that e itself, and not some intermediate rendering of it, is always available to compare against. That assumption is false for essentially every substrate chain of practical interest, and its falsity is precisely what Chapter 5 identified as the reason the repair-budget appa-

ratus of Chapter 4 does not, by itself, resolve the continuation problem of Chapter 1.

6.2 The Substrate Chain

Definition 6.1 (Substrate Chain). A substrate chain is a sequence

$$A_0 \xrightarrow{B_0} A_1 \xrightarrow{B_1} \dots \xrightarrow{B_{n-1}} A_n$$

where each A_i is a rendering (definition 2.2) and each transition is governed by a local budget triple $B_i = (C_i, I_i, \beta_i)$: a local repair cost function, a local interference function, and the budget available at that stage.

Two restrictions are built into definition 6.1 and are worth stating openly rather than leaving implicit. First, the chain is linear and finite: each A_i has exactly one predecessor and one successor, and there are finitely many stages. The branching generalization, in which a single A_i gives rise to several independent successors, is deferred to Chapter 8, where it is shown to introduce a genuinely new failure mode (silent cross-branch disagreement) rather than being a straightforward extension of what follows here. Whether a substrate chain with infinitely many stages behaves differently in kind, and not merely in degree, from the finite case treated throughout this monograph is left unaddressed and is recorded as an open question in Appendix F. Second, A_0 is not assumed to be the original object e of Chapter 2; it is only assumed to be the earliest rendering the chain's analysis takes as given. A substrate chain is therefore always relative to a choice of starting point, and nothing in this chapter's machinery requires that starting point to be loss-free relative to whatever preceded it.

6.2.1 Local Budgets as Stage-Local Repair Economies

The triple $B_i = (C_i, I_i, \beta_i)$ is not an unrelated new primitive. It is Chapter 4’s repair-economic apparatus, restricted to a single stage and to that stage’s own repair-stable domain. Concretely, C_i is the restriction of the repair cost function $C_R(\cdot)$ of definition 4.2 to the distinctions present at stage i ; I_i is the restriction of the interference functional $I(\cdot, \cdot)$ of definition 4.5 to pairs of distinctions competing for stage i ’s budget; and β_i plays the role, at stage i alone, that the total budget Φ_{tot} plays for a single observer in theorem 4.6. The relationship between β_i and Φ_{tot} is best understood by contrast rather than by analogy: Φ_{tot} is conserved across an observer’s entire, simultaneous distinction space, so that theorem 4.6 can speak of resources being redistributed *within* it. No such global quantity is conserved across a substrate chain’s stages. $\beta_0, \beta_1, \dots, \beta_{n-1}$ are, in general, entirely independent numbers set by independent parties for independent reasons, and nothing in definition 6.1 relates β_i to β_j for $i \neq j$. This is the formal expression of the distinction drawn informally in Chapter 5: the Redistribution Theorem governs one Φ_{tot} shared by many distinctions; a substrate chain has many β_i , each shared by nothing beyond its own stage.

6.3 Budget-Constrained Discard, and Its Underdetermination

Definition 6.2 (Budget-Constrained Discard). A distinction d present in A_i is dropped in A_{i+1} if maintaining d exceeds β_i , or if maintaining d interferes too strongly, per I_i , with a higher-priority distinction competing for the same stage’s budget.

definition 6.2 is deliberately schematic, and it is worth being honest, before building anything further on it, about exactly what the schema leaves open. The phrase “higher-priority distinction” presupposes a priority ordering over the distinctions

present at stage i , and definition 6.2 does not say where that ordering comes from or what happens when two admissible orderings would license different discards under the same budget β_i . This is not a minor omission to be patched in passing. Two natural instantiations of definition 6.2 exist, they are not equivalent, and the monograph's later theorems need to be checked against both rather than silently assumed to hold for whichever one is easiest to compute with.

Construction 6.3 (Greedy Priority Discard). Fix a priority ordering $d_{(1)} \succ d_{(2)} \succ \cdots \succ d_{(m)}$ over the distinctions present at stage i , reflecting their importance to the stage's own task τ_i . Process the distinctions in this order, maintaining a running total of committed budget. A distinction is funded, and hence preserved into A_{i+1} , if the remaining budget after funding every higher-priority distinction is at least $C_i(d)$; otherwise it is dropped, and processing continues to the next distinction in the ordering without returning to reconsider a dropped distinction later.

Construction 6.4 (Cost-Optimal Discard). Given the same priority ordering interpreted now as a value assignment $v(d_{(k)})$ decreasing in k , select the subset S of stage i 's distinctions maximizing $\sum_{d \in S} v(d)$ subject to $\sum_{d \in S} C_i(d) \leq \beta_i$, breaking ties in favor of higher-priority distinctions.

Remark 6.5 (The Two Constructions Diverge). construction 6.3 and construction 6.4 are the discard rule and the knapsack-optimal packing of the same underlying priorities and costs, respectively, and they can select different subsets for the same (C_i, I_i, β_i) and the same priority ordering. A single low-priority, low-cost distinction that the greedy rule would skip in order to fund several higher-priority items may be exactly the distinction a global optimization would sacrifice one mid-priority item to include, whenever doing so raises the total value funded. Neither construction is privileged by definition 6.2 as stated;

the definition is a constraint on the *outcome* of a stage’s discard decision, not a specification of the *procedure* that produces it.

This divergence is not a defect to be resolved before the theory can proceed. It is the correct level of generality for a schema meant to cover discard decisions made by a codec’s rate-control algorithm, a human editor’s judgment, a court’s interpretation of relevance, and an institution’s retention policy, none of which can be assumed to implement the same procedure. What it does mean, and what is recorded explicitly here so that it cannot be forgotten by the time Appendix A is drafted, is that any theorem claiming to hold for *every* budget relay — including the Composition Failure Theorem of Chapter 9 — must hold under both construction 6.3 and construction 6.4, and ideally under any construction satisfying only the qualitative constraint of definition 6.2. This robustness requirement is added to Appendix F as a condition Appendix A’s eventual proof must be checked against, not assumed to satisfy by construction.

6.4 The Budget Relay, Defined

Definition 6.6 (Budget Relay). A substrate chain (definition 6.1) is a *budget relay* if, for every stage i , the transition B_i is selected using only the current rendering A_i and the stage’s own local task τ_i — that is, without reference to the loss ledgers $L_0, \dots, L_{i-1}$ of any earlier stage, without reference to the budget triple B_j of any other stage $j \neq i$, and without appeal to any interference term spanning distinctions held at different stages.

This restates, and sharpens, an earlier and vaguer formulation of the same condition. The earlier formulation spoke of stages having no “shared interference term or shared discard record...by default,” which left unclear exactly what a stage is

and is not permitted to consult when setting its own budget. definition 6.6 as now stated answers that question directly: a stage may see the rendering handed to it and may know its own task, and nothing else about the chain’s history or future. This is doing the real work of the definition, and it is worth dwelling on why. A substrate chain in which every stage consults a common ledger and a common, chain-wide interference functional before setting its own budget is, for the purposes of this monograph’s apparatus, indistinguishable from a single observer’s economy distributed across time: its total behavior is governed by theorem 4.6 rather than by anything new. That case is not excluded from consideration — Chapter 7 treats exactly this kind of shared-ledger access as one point on a spectrum of partial-coordination mechanisms — but it is not the default case, and definition 6.6 is written to isolate the default case cleanly: a relay in which coordination, if it exists at all, must be added back in deliberately and explicitly, rather than assumed from the outset.

Remark 6.7 (Degenerate Cases). A substrate chain with $n = 1$ satisfies definition 6.6 vacuously, since there is only one stage and no other stage’s budget or ledger to consult; this case reduces to the single rendering already treated in Chapters 2–3 and adds nothing new. The genuinely new content of definition 6.6 appears only once $n \geq 2$, and the sharpest illustrations of that content, developed in section 6.8 below, use $n = 3$.

6.5 The Root-Object Move

proposition 3.7 showed that sufficiency cannot be certified at the time a substrate is produced, because the relevant future task space is generally unknown then. Within a budget relay this observation sharpens into a design question rather than remaining a mere impossibility result: since no stage can know which of its discards a future, unanticipated task will need

reversed, is there anything a relay’s earliest stage can do that does not require anticipating the future task space at all?

Definition 6.8 (Root Object). A distinguished rendering A_0 in a substrate chain, treated as the reference point against which all downstream loss is measured. Where possible, ingestion-time policy should optimize for re-derivability of downstream renderings from A_0 , rather than for sufficiency relative to an as-yet-unknown τ .

Proposition 6.9 (Root Recovery). *If the root object A_0 survives unaltered, any downstream rendering A_i can be regenerated under an alternative budget allocation B'_i .*

Proof Sketch. By definition 6.1, every A_i for $i > 0$ is generated from A_0 via some composition of transitions. If A_0 persists, the transition B_i can be replaced with B'_i and A'_i recomputed directly from A_0 ; the loss incurred under B_i is therefore reversible whenever A_0 is available. Without a preserved A_0 , distinctions discarded at any stage are permanently unavailable to every later stage, since definition 6.2 provides no mechanism for a distinction to reappear once dropped.

proposition 6.9 is easy to overstate, and it is worth being precise about what it does not claim. It does not claim that preserving A_0 is free. Preserving a root object is itself a repair decision – A_0 must be maintained against whatever perturbations threaten it (storage degradation, format obsolescence, simple deletion under retention-policy pressure) – and that maintenance itself has a cost that must be funded by *some* budget, decided by *someone*, before any downstream task requiring it is known to exist.

Remark 6.10 (The Root-Preservation Bootstrapping Problem). Preserving A_0 so that proposition 6.9 remains available is itself an instance of proposition 3.7: the decision to fund A_0 ’s preservation must be made before the future task space that would

justify the expense is known, by definition of that task space being open. An institution cannot, even in principle, calculate the return on archiving a root object against tasks that do not yet exist. This does not undermine definition 6.8's recommendation so much as clarify its status: root preservation is not a solution to the sufficiency problem of chapter 3, but a wager that re-derivability will be worth more, in expectation, than whatever the preservation itself costs. Whether this wager can be given a more precise decision-theoretic treatment, rather than being left at the level of a design recommendation, is left open in Appendix F.

6.6 Cumulative Discard and Its Monotonicity

The claim that loss compounds along an uncoordinated chain has been used informally since Chapter 5. Making it precise requires first defining what “cumulative” discard means for a chain of more than one transition, since definition 2.5 was stated only for a single rendering.

Definition 6.11 (Cumulative Discard Set). For a substrate chain and indices $i \leq j$, write $D_k := D(A_k \rightarrow A_{k+1})$ for the discard set of the single transition B_k . The cumulative discard set between stages i and j is

$$D(A_i \Rightarrow A_j) := \bigcup_{k=i}^{j-1} D_k.$$

Proposition 6.12 (Monotonicity of Cumulative Discard). *For any substrate chain in which the root object is not available for recovery (proposition 6.9 does not apply), and for any $i \leq j \leq l$,*

$$D(A_i \Rightarrow A_j) \subseteq D(A_i \Rightarrow A_l).$$

Proof Sketch. Immediate from definition 6.11: $D(A_i \Rightarrow A_l)$ is a union over a strictly larger index range than $D(A_i \Rightarrow A_j)$, and a union over a superset of index terms contains the union over the subset. No further argument is required once cumulative discard is defined as a union of per-transition discard sets, since definition 6.2 supplies no operation by which a distinction can be re-added to a later A_l once absent from an intermediate A_k for $k < l$.

This result was labeled a Theorem in an earlier draft of this chapter, and it is worth recording explicitly why that label has been withdrawn. proposition 6.12’s inequality is, once definition 6.11 is stated correctly, an immediate consequence of set union rather than a substantive derivation, and the monograph’s own epistemic-labeling discipline reserves the word “Theorem” for results whose proof carries independent weight. The genuine content of the claim that loss compounds along a chain is not the inclusion itself but the prior, harder-won observation that definition 6.2 provides no mechanism for reversal absent proposition 6.9 – and that observation is a modeling commitment built into definition 6.1, not a theorem about it.

6.7 The Loss Ledger, Upgraded

Definition 6.13 (Budget-Aware Loss Ledger). The loss ledger L_i attached to transition B_i records not merely the discard set $D_i = D(A_i \rightarrow A_{i+1})$, but the pair $L_i = (D_i, B_i)$: the discarded distinctions *together with* the local budget that produced the discard. This allows a downstream consumer to distinguish a distinction dropped because it was irrelevant to that stage’s own purpose from one dropped because that stage’s budget was simply small – a distinction with different implications for whether the loss might be recoverable elsewhere in the relay.

definition 6.13 is what separates a budget relay’s provenance machinery from the plain loss ledger of Chapter 2: it

is a coordination device, not merely a passive record, and its role as such is developed fully in Chapter 7. It is worth noting immediately, however, a limitation that definition 6.6 itself imposes on L_i 's usefulness: because a budget relay's stage $i + 1$ does not consult L_j for $j < i$ when setting B_{i+1} , the mere existence of a loss ledger at every stage does not, by itself, prevent the composition failures analyzed in Chapter 9. A loss ledger is a necessary condition for a downstream consumer, arriving after the fact and inspecting the whole chain, to diagnose what happened; it is not, on its own, a mechanism by which an intermediate stage avoids causing the problem in the first place.

6.8 A Fully Worked Instance

The definitions above are abstract enough that their consequences are easy to state and easy to misjudge. This section fixes one small, fully explicit numerical relay and works every definition above against it by hand, so that later chapters – particularly the proof of Chapter 9 and its constructive witness in Chapter 20 – have a concrete instance to generalize from rather than only an abstract schema.

Consider a source rendering A_0 carrying five distinctions, $d_1, \dots, d_5$, with repair costs $C(d_1) = 1$, $C(d_2) = 2$, $C(d_3) = 3$, $C(d_4) = 2$, and $C(d_5) = 4$, and suppose interference among them is negligible, so that $I \approx 0$ throughout and the greedy construction (construction 6.3) and the optimal construction (construction 6.4) can be compared directly against the same numbers.

6.8.1 Stage $0 \rightarrow 1$

Suppose stage 0's task τ_0 ranks the distinctions by priority $d_1 \succ d_2 \succ d_4 \succ d_5 \succ d_3$, and that $\beta_0 = 4$. Applying con-

struction 6.3: d_1 is funded (cost 1, remaining budget 3); d_2 is funded (cost 2, remaining budget 1); d_4 cannot be funded (cost 2 > 1 remaining) and is dropped; d_5 cannot be funded (cost 4 > 1 remaining) and is dropped; d_3 cannot be funded (cost 3 > 1 remaining) and is dropped. Thus A_1 retains $\{d_1, d_2\}$ and $D_0 = \{d_3, d_4, d_5\}$. Applying construction 6.4 to the same priorities and budget yields the identical subset in this instance, since $\{d_1, d_2\}$ is in fact the value-maximizing feasible set here; remark 6.5's divergence is real in general but does not arise for every choice of numbers, and this instance was chosen partly to establish a clean baseline before a divergent case is considered elsewhere.

The loss ledger at this stage is $L_0 = (\{d_3, d_4, d_5\}, B_0)$, with B_0 recording $\beta_0 = 4$ against the priority ordering above – so that a downstream consumer inspecting L_0 can see that d_3, d_4 , and d_5 were dropped because τ_0 ranked them low and β_0 was small relative to their combined cost, not because they were judged worthless in any absolute sense.

6.8.2 Stage 1 $\rightarrow$ 2

Suppose stage 1's task τ_1 requires only $\{d_1, d_2\}$, exactly what survived into A_1 , and that $\beta_1 = 3$ – exactly sufficient to fund both at cost 1 + 2 = 3. Then A_2 retains $\{d_1, d_2\}$ and $D_1 = \emptyset$. By definition 6.11, $D(A_0 \Rightarrow A_2) = D_0 \cup D_1 = \{d_3, d_4, d_5\}$, unchanged from $D(A_0 \Rightarrow A_1)$ – an explicit instance of proposition 6.12's inclusion, here realized as equality rather than strict containment, since stage 1 happened to discard nothing further.

6.8.3 A Task Neither Stage Anticipated

Suppose a task τ^* is later discovered, requiring recovery of d_3 – some distinction, present in A_0 , that neither τ_0 nor τ_1 had reason to prioritize. Since $d_3 \notin A_1$ and A_2 is derived only from A_1 ,

d_3 is unrecoverable from A_2 by definition 6.2's irreversibility and confirmed unrecoverable in general by proposition 6.12. Neither B_0 nor B_1 was irrational relative to its own task: β_0 was spent optimally against τ_0 's stated priorities, and β_1 was spent optimally, indeed exactly, against τ_1 's. A_2 is nonetheless insufficient (definition 3.5) for τ^* . This four-paragraph construction is already, in miniature, the mechanism that Chapter 9 formalizes as Theorem 9.2 and that Chapter 20 exhibits again at full scale against a real media pipeline; nothing about the general theorem's proof will do more than what has already happened here with five distinctions and two stages.

6.8.4 Root Recovery, Demonstrated

Suppose, finally, that A_0 itself was preserved as a root object (definition 6.8) rather than discarded once A_1 was produced. Upon discovering τ^* , a new transition B'_0 can be computed directly from A_0 using a revised priority ordering that ranks d_3 above d_4 , yielding a new rendering A'_1 retaining $\{d_1, d_2, d_3\}$ and satisfying τ^* . This is proposition 6.9 made concrete: the insufficiency exhibited above was permanent for the original chain $A_0 \rightarrow A_1 \rightarrow A_2$, but not permanent for the underlying situation, precisely because A_0 , and not merely A_1 or A_2 , remained available to be re-derived from.

6.9 What This Definition Commits Us To

It is worth closing this chapter with an explicit account of exactly what has been assumed, so that later chapters relaxing any one of these assumptions can be read as deliberate extensions rather than silent changes of subject. A substrate chain, as defined here, is linear rather than branching (relaxed in Chapter 8); finite rather than unbounded (left open in Appendix F); governed at each stage by a discard rule whose outcome, but

not procedure, is constrained by definition 6.2 (with the resulting underdetermination flagged in remark 6.5 as a robustness condition the proofs of Chapter 9 must satisfy); irreversible in its losses except through the specific, costly mechanism of root recovery (proposition 6.9, itself subject to the bootstrapping problem of remark 6.10); and uncoordinated by default, in the precise sense given in definition 6.6, with coordination treated in Chapter 7 as something that must be added back in deliberately rather than assumed. Every one of these five commitments is a modeling choice, not a discovered fact about the world, and the worked instance of section 6.8 was constructed specifically to make each commitment's consequences visible in a case small enough to check by hand before trusting the general theory built on top of it.

Chapter 7

Coordination, Its Absence, and Partial Remedies

Synopsis. This chapter investigates what happens when stages within a relay attempt to compensate for the absence of global coordination. Augmentation, provenance, priors, adjudication, and revision are reinterpreted not as an unrelated list of techniques but as distinct strategies for partially coordinating otherwise independent repair economies, each addressing a different failure mode: augmentation where hidden state is enumerable, provenance where loss can at least be recorded, priors where a consumer can plausibly fill a benign gap, adjudication where authority can terminate ambiguity without recovering information, and revision where the relay itself must be replaced. None fully solves the coordination problem; each narrows a specific failure mode.

7.1 Why Full Coordination Is Usually Unavailable

A field-recording engineer, the vendor of a transcription model, and a human editor who later summarizes the transcript typically share no protocol, no common budget, and often no direct communication at all. definition 6.6’s uncoordinated case is not a pathological edge case to be assumed away; for most substrate chains of practical interest, it is the default condition. This chapter asks what a stage *can* do to compensate for that default, given that Chapter 5 already showed (proposition 5.1) that no extension of Chapter 4’s single-observer apparatus is available to fall back on.

Definition 7.1 (Coordination Mechanism). A coordination mechanism is any deliberate modification to definition 6.6’s default independence condition, permitting some stage or external party access to information — a loss ledger, a hidden-state enumeration, an authoritative ruling, or the raw root object itself — that a fully uncoordinated relay would not provide.

Five such mechanisms recur across the domains treated in Part IV, and it is worth resisting the temptation to treat them as an unrelated list of ad hoc devices collected because they happen to be useful. Each addresses a specific, identifiable gap in definition 6.6’s independence condition, and organizing them by which gap they close is more informative than organizing them alphabetically or by domain.

7.2 Augmentation

Augmentation adds an enumerated hidden-state set H directly to a rendering, pre-committing budget to distinctions a stage’s own task does not itself require, on the strength of knowing in advance that some later consumer will. By proposition 3.10,

this strategy is viable only in closed hidden-state domains (definition 3.8): Chapter 13 develops the case in full, where H — castling rights, en passant status, repetition count — is finite and known at the time a position is recorded, so that funding it requires no prediction about which future task will need it, only the (comparatively modest) foresight that *some* task among a known, finite list might.

Remark 7.2 (Augmentation Does Not Solve the Open Case). Augmentation is unavailable, not merely difficult, in open hidden-state domains (definition 3.8): there is no finite H to pre-commit budget to, since proposition 3.10 shows no finite augmentation closes an open domain’s hidden-state space. Painting (Chapter 14) is not a case where augmentation would help if only more budget were available; it is a case where augmentation, as a strategy, does not apply regardless of budget.

7.3 Provenance and Loss-Ledger Sharing

Definition 7.3 (Provenance-Coordinated Relay). A substrate chain is provenance-coordinated if definition 6.6’s independence condition is relaxed to permit stage i ’s transition B_i to be selected with reference to the loss ledgers $L_0, \dots, L_{i-1}$ of every earlier stage, while still forbidding reference to any later stage’s budget or ledger.

This is the weakest relaxation of definition 6.6 that still qualifies as coordination at all: it requires no advance agreement between stages, only that each ledger be produced and made accessible to whatever comes after it, and it preserves the chain’s directionality — a stage still cannot see the future, only more of the past than definition 6.6 otherwise allows.

Proposition 7.4 (Provenance Sharing Does Not Restore Sufficiency Alone). *A provenance-coordinated relay (definition 7.3) can still exhibit the failure mode of theorem 9.2.*

Proof Sketch. Reading $L_0, \dots, L_{i-1}$ tells stage i , and eventually a downstream consumer, *what* was discarded and under what budget — but not what a future, unknown τ^* will require, since proposition 3.7 applies to the ledger’s reader exactly as it applies to the stage that produced the ledger. Knowing that d^* was dropped at stage 0 does not, by itself, supply d^* ; recovering it requires either proposition 6.9’s root access, which definition 7.3 does not grant, or advance knowledge of τ^* , which proposition 3.7 rules out in general. Provenance sharing converts an undetectable insufficiency into a diagnosable one; it does not convert it into an avoided one.

proposition 7.4 is why Chapter 6 described the budget-aware loss ledger (definition 6.13) as a coordination device rather than a solution: it is necessary for a downstream party to *diagnose* what happened, which matters a great deal for the reasons Chapter 20 develops, but it is not sufficient to *prevent* what happened, which requires the stronger and costlier machinery of root preservation.

7.4 Priors

A consumer-side substitute for coordination: when a relay supplies no interference information, the consumer supplies plausible reconstructions of the missing distinction from general knowledge. Painting (chapter 14) is the paradigm case, and it is available precisely because, and only because, the discarded distinction’s absence is benign (definition 2.7) relative to the consumer’s task: the correctness set $\mathcal{C}_\tau(e)$ of definition 3.4 is large enough that a plausible guess, rather than the exact discarded distinction, is sufficient to land inside it. Priors are unavailable, or worse than unavailable, exactly where definition 2.8’s conditions hold: a confidently wrong prior substituted for an undisclosed loss is corruption compounded by false confidence, not a benign completion.

7.5 Adjudication

Proposition 7.5 (Adjudication Terminates Ambiguity). *Let $\mathcal{J}$ be an authority external to a substrate chain. Adjudication by $\mathcal{J}$ produces a unique continuation even when the distinguishability information available in the chain is incomplete.*

Proof Sketch. Under adjudication, the continuation is selected by an authoritative decision procedure rather than inferred from available evidence. The underlying information deficit — the discarded distinctions — is unaffected by the ruling; what changes is that ambiguity is terminated procedurally rather than resolved informationally.

It is worth stating precisely, in the vocabulary of Chapter 3, what adjudication actually does to definition 3.4’s correctness set. Adjudication does not discover which continuation truly belongs to $\mathcal{C}_\tau(e)$; it substitutes $\mathcal{J}$ ’s ruling as the operative correctness set going forward, regardless of whether that ruling coincides with whatever the undiscoverable true $\mathcal{C}_\tau(e)$ would have contained. Chapter 17 treats this substitution as the defining feature of legal continuation under irreducible uncertainty: courts do not typically claim to have recovered the parties’ true intent so much as to have settled, bindingly, what will be treated as though it were. Adjudication is, in this taxonomy, the only mechanism that terminates ambiguity without requiring any additional distinguishability information to be recovered or supplied. It trades informational completeness for a binding decision.

7.6 Revision

Definition 7.6 (Relay Revision). Wholesale replacement of an entire substrate chain $A_0 \rightarrow \cdots \rightarrow A_n$ by a new chain built from new production, rather than re-derived from the original chain’s root object.

Revision is easy to confuse with root recovery (proposition 6.9), and it is worth being precise about the difference. Root recovery re-derives a new downstream rendering A'_i from the *same* preserved A_0 , under a revised discard policy; it presupposes A_0 survived and is available. Revision, as defined here, does not presuppose this: it discards the entire existing chain, root included, and begins again from new evidence, a new theory, or a new production process entirely. Chapter 18 develops this case in full: a superseded scientific theory is not re-derived from some preserved “root” observation set in the way proposition 6.9 re-derives a rendering; the theory itself, and often the observational framework that produced the evidence for it, is replaced.

7.7 Coordination Versus Pooling: A Boundary Case

The five mechanisms above all operate *within* the assumption that stages remain separate — each retains its own local budget, and coordination consists of granting limited, structured access across that separation (a ledger here, an authority there), not erasing it. Appendix C’s demonstration that pooling the witness’s stage budgets into one shared total can restore sufficiency (proposition C.1) is a more radical move than any mechanism in this chapter’s taxonomy: pooling does not coordinate separate local economies, it eliminates the separation itself, collapsing the relay back into the single-observer setting Chapter 4 already covers. None of augmentation, provenance-sharing, priors, adjudication, or revision does this; each leaves definition 6.1’s stage structure intact and works within it. This is worth flagging explicitly because pooling is, in a precise sense, strictly more powerful than anything on this chapter’s list — it is what makes theorem 4.6 available again — and its absence from the taxonomy is not an oversight but a reflection

of how rarely it is actually achievable: pooling requires exactly the kind of unified, advance-negotiated authority over an entire chain's resources that Section 7.2's discussion of augmentation already identified as the exception rather than the rule.

7.8 The Limits of the Taxonomy

Proposition 7.7 (No Coordination Mechanism Restores Universal Sufficiency). *No coordination mechanism drawn from Sections 7.2–7.6 can guarantee sufficiency of a substrate chain over an open task space T .*

Proof Sketch. This follows directly from proposition 3.7: each coordination mechanism above operates by preserving, recovering, or authoritatively resolving *some additional* set of distinctions, but T 's openness means a task τ^* can always be constructed requiring a distinction outside whatever finite additional set any given mechanism supplies. Coordination narrows the gap identified in principle 5.2; it does not close it. This result was labeled a Theorem in an earlier draft; the label has been corrected to Proposition here, consistent with the monograph's epistemic-labeling discipline, since the argument is a direct application of proposition 3.7 rather than an independent derivation.

Chapter 9 is, in effect, the sharper and more constructive version of proposition 7.7: it does not merely state that no mechanism guarantees universal sufficiency, but exhibits the mechanism by which locally rational, uncoordinated stages compose into global insufficiency in the first place, and Appendix A exhibits it with a construction robust to the choice of local discard rule.

CHAPTER 7. COORDINATION, ITS ABSENCE, AND PARTIAL REMEDIES

Chapter 8

Fan-Out: Parallel Budget Relays and Cross-Branch Disagreement

Synopsis. The framework is extended here from linear chains to branching structures, where a single source generates multiple downstream renderings optimized for different purposes. Branch agreement and silent disagreement are introduced to describe situations in which branches appear compatible while relying on incompatible assumptions about discarded information. Confidence and quality estimates are shown to be renderings in their own right, subject to the same limitations as the artifacts they describe, which motivates the chapter's closing move: a trust boundary as the practical stopping point that prevents an infinite regress in evaluating confidence and provenance systems.

8.1 From Chains to Graphs

Definition 8.1 (Fan-Out Relay). A fan-out relay is a directed graph of renderings with a single source A_0 and two or more independent successor branches $A_0 \xrightarrow{B^{(1)}} A^{(1)}$, $A_0 \xrightarrow{B^{(2)}} A^{(2)}$, ..., each governed by its own local budget triple and each satisfying definition 6.6’s independence condition with respect to every other branch: branch k ’s transition is selected without reference to any other branch’s budget or ledger.

definition 8.1 generalizes definition 6.1 directly: audio rendered in parallel into a transcript, an acoustic-feature analysis, and an emotion analysis is the paradigm case, and each branch may itself continue as a linear substrate chain of arbitrary length beyond the point of divergence. What is new here is not the branching structure by itself, but what it makes possible: two renderings that both trace back to the same source, and that a downstream party might reasonably expect to agree with each other, can instead diverge in ways that neither branch’s own local admissibility (definition 9.1) does anything to prevent.

8.2 Branch Agreement and Silent Disagreement

Definition 8.2 (Branch Agreement). Two renderings $A^{(i)}$, $A^{(j)}$ derived from a common ancestor are in agreement on a distinction d if neither branch’s discard set contradicts the other’s claims about d .

Definition 8.3 (Silent Disagreement). Branches disagree silently when each branch’s confidence or quality profile Q fails to flag that another branch depended on a distinction that the first branch discarded or rendered unstable.

The definitions above are easy to accept in the abstract and easy to under-examine in practice, so this section works a single construction in full, in the same style as Appendix A's canonical witness, rather than resting on a description of the transcript/acoustic case alone.

8.2.1 A Minimal Two-Branch Witness

Construction 8.4 (Two-Branch Witness). Let A_0 carry two distinctions: d , the resolved identity of an ambiguous phoneme, with cost $C(d) = 2$; and q_d , a meta-distinction recording whether the source evidence for d is stable or unstable, with cost $C(q_d) = 1$. Branch X (a transcript) has task τ_X requiring exactly $\{d\}$, with budget $\beta_X = 2$. Branch Y (an acoustic-stability analysis) has task τ_Y requiring exactly $\{q_d\}$, with budget $\beta_Y = 1$.

Proposition 8.5 (The Witness Produces Silent Disagreement). *In construction 8.4, branch X is locally admissible with respect to τ_X and discards q_d ; branch Y is locally admissible with respect to τ_Y and discards d ; and a consumer with access only to branch X cannot detect that q_d – and hence the instability branch Y would have reported – was ever at issue.*

Proof. For branch X : the required distinction $\{d\}$ has cost $R_X = C(d) = 2 = \beta_X$ exactly. The only non-required distinction present, q_d , has cost $C(q_d) = 1 > \beta_X - R_X = 0$. By lemma A.4, q_d is dropped under any construction satisfying definition 6.2 with d ranked above q_d , so X is locally admissible (definition 9.1) and $D_X = \{q_d\}$.

For branch Y : symmetric. The required distinction $\{q_d\}$ has cost $R_Y = C(q_d) = 1 = \beta_Y$ exactly, and the non-required d has cost $C(d) = 2 > \beta_Y - R_Y = 0$, so by lemma A.4, d is dropped under any construction, Y is locally admissible, and $D_Y = \{d\}$.

By proposition 8.7, X 's confidence profile Q_X is itself a rendering, produced from whatever X retains; since $q_d \notin A^{(X)}$,

Q_X has no distinction available from which to construct an instability flag, and X 's output presents d 's resolved value without qualification. A consumer relying on branch X alone therefore receives no signal — not a low-confidence flag, not a declared gap, nothing — indicating that q_d was ever a live consideration, even though branch Y , derived from the identical source A_0 , would have reported it as unstable. This is exactly definition 8.3's condition: Q_X fails to flag that another branch's finding (instability, via q_d) bears on distinction d , which X itself depended on. $\square$

This construction is deliberately built to reuse Appendix A's exhaustion lemma rather than argue the point freshly, and the reuse is informative in its own right: the same mechanism that forces a construction-independent discard within a single linear chain (lemma A.4) forces construction-independent, *asymmetric* discard across two branches of a fan-out, and it is precisely this asymmetry — X drops what Y keeps, and vice versa — that produces silent disagreement rather than mere independent loss.

Proposition 8.6 (Silent Disagreement Exists in General). *Given two branches produced independently (definition 8.1) from a common source, there exist source distinctions whose instability is reported by one branch and concealed by the other.*

Proof Sketch. construction 8.4 together with proposition 8.5 constitutes an existence proof: at least one pair of branches, with at least one such distinction, is exhibited explicitly and shown to satisfy definition 8.3 under a construction-independent argument. This result was labeled a Theorem in an earlier draft; the label has been corrected to Proposition, consistent with the monograph's epistemic-labeling discipline, since an existence claim witnessed by one explicit construction is not of the same weight as theorem 9.2's general composition result.

8.3 The Regress Inside the Confidence Profile

Proposition 8.7 (Confidence Profiles Are Renderings). *A confidence or quality profile Q attached to a rendering A is itself produced by a projection, and therefore has its own discard set $D(Q)$.*

Proof Sketch. Confidence estimation maps source evidence – acoustic likelihoods, model logits, annotator judgments – into a finite label or score. Any such finite mapping collapses distinctions in the sense of definition 2.2; Q therefore satisfies the definition of a rendering and inherits a discard set like any other.

proposition 8.7 produces a regress: trusting Q requires trusting the process that produced Q , which has its own confidence measure Q' , and so on. This regress does not resolve within the formalism; it is cut off by convention rather than by argument, and it is worth being explicit about the cost of extending it even one level further. Verifying Q against a further Q' is itself a repair decision in the sense of Chapter 4: it consumes budget that could otherwise be spent on the object-level distinctions Q was meant to qualify, and proposition 5.1's argument applies just as well to a chain of nested confidence measures as to a chain of substantive renderings. Real systems stop extending this chain not because the regress terminates on its own but because someone decides, by convention, where it is no longer worth the expenditure.

Definition 8.8 (Trust Boundary). A designated layer – for instance, a fixed, versioned confidence model – beyond which a relay does not recursively question its own loss-accounting. The trust boundary is a declared convention, not a theorem; proposition 8.7's regress is halted by stipulation at this layer, and an auditor of the relay is, in effect, trusting the boundary itself rather than anything beyond it.

Definition 8.9 (Boundary-Relative Sufficiency). A rendering is boundary-sufficient for a task if it is sufficient (definition 3.5) conditional on every confidence and quality profile at or beyond the declared trust boundary (definition 8.8) being accepted without further verification.

definition 8.9 names what practitioners actually rely on, as distinct from the stronger and generally unattainable notion of unconditional sufficiency: a system is almost never verified all the way down to its source distinctions, and boundary-sufficiency makes explicit exactly how much trust that verification is quietly resting on.

Chapter 9

The Composition Failure Theorem

Synopsis. This chapter presents the monograph's central new result. The Composition Failure Theorem demonstrates that a sequence of locally rational transformations can collectively produce a globally insufficient outcome, with no stage behaving negligently, incompetently, or irrationally: failure emerges because each stage optimizes for its own local task while future tasks remain unknown. This formalizes a pattern familiar from archival systems, institutions, scientific communication, and information-processing pipelines, in which a distinction that seemed irrelevant at one stage becomes essential later.

(New to this monograph.)

9.1 Making Local Admissibility Precise

Before the theorem can be stated, the phrase it turns on — a transition being “locally admissible with respect to its own stage’s task” — needs a definition sharper than definition 6.2’s general schema alone provides. definition 6.2 says a distinction is dropped when funding it exceeds budget or when it interferes too strongly with a higher-priority distinction; it does not say how priority itself is supposed to track a stage’s task. The following definition closes that gap, and everything downstream — the theorem, its corollaries, and the proof of Appendix A — depends on exactly this closure.

Definition 9.1 (Locally Admissible Transition). A transition B_i is locally admissible with respect to τ_i if its discard set D_i is produced by a construction satisfying definition 6.2 in which every distinction required by τ_i (definition 2.7) is ranked strictly above every distinction not required by τ_i .

This definition is deliberately weak in one respect and strong in another. It is weak in that it does not choose between construction 6.3 and construction 6.4, or any other construction satisfying definition 6.2: any of them counts as locally admissible provided the priority ordering they operate on is correctly aligned with τ_i . It is strong in that this alignment is not optional — a transition that funds a distinction τ_i does not need ahead of one it does need fails to be locally admissible, regardless of how sophisticated the underlying discard construction is. A transition satisfying definition 9.1 cannot be accused of irrationality relative to τ_i : it funds everything τ_i needs, in full, before spending anything on distinctions τ_i does not need. This is exactly the standard the theorem needs every stage to meet, since the entire point of the result is that failure can occur even when no stage falls short of it.

9.2 Statement

Theorem 9.2 (Composition Failure). *Let $A_0 \xrightarrow{B_0} \dots \xrightarrow{B_{n-1}} A_n$ be a budget relay (definition 6.6) in which each transition B_i is locally admissible (definition 9.1) with respect to its own stage's task τ_i . Then there exist open task spaces T (proposition 3.7) containing some $\tau^* \notin \{\tau_0, \dots, \tau_{n-1}\}$ such that A_n is insufficient (definition 3.5) for τ^* , even though no individual transition was locally irrational relative to its own τ_i .*

Each clause of this statement is load-bearing, and it is worth checking what happens if any one is dropped. Without definition 6.6's independence condition, the chain reduces to a single observer's economy and theorem 4.6 governs it instead, as Appendix C shows concretely. Without local admissibility, the theorem would be uninteresting: of course an incompetently-run relay can end up insufficient. Without T being open, τ^* could in principle have been anticipated and funded for, and the failure would trace back to ordinary negligence rather than to the structure of uncoordinated composition itself. The theorem's content is precisely that none of these outs are available: the relay is uncoordinated by definition, every stage did what it was supposed to do, and the task that breaks it was not available to be planned for.

9.3 Proof Strategy

The full proof is given in Appendix A; this section previews its shape so the construction does not appear unmotivated when read there.

Proof Sketch. The proof constructs an explicit witness relay using the five-distinction instance introduced in section 6.8 and reused in section 4.5, together with a family of *recovery tasks* τ_d — one for each named distinction d , asking only whether

d holds in the source (definition A.1) — which serves as the open task space T . A stage whose required distinctions exactly, or near-exactly, exhaust its local budget is shown (lemma A.4) to drop every non-required distinction *regardless of which construction satisfying definition 6.2 is used*, which is what allows the witness to be locally admissible in the strong sense of definition 9.1 without leaving room for a more clever discard policy to have rescued the dropped distinction. Some distinction d^* — concretely, d_5 in the worked witness — is required by neither stage’s task and is dropped at the earliest stage as a direct consequence. By proposition 6.12, d^* remains unavailable at every later stage absent root recovery (proposition 6.9), which the witness does not invoke. Taking $\tau^* = \tau_{d^*}$ then gives insufficiency directly, by lemma A.2: a rendering is sufficient for a recovery task exactly when it has not discarded the distinction in question, and A_n has.

This preview differs in one respect from an earlier, less precise sketch of the same argument, which described d^* as dropped because it “interferes with a distinction τ_i does require.” The actual mechanism proved in Appendix A is sharper and does not rely on interference at all: d^* is dropped because, after funding every required distinction in full, no budget capable of funding d^* remains — a conclusion that holds regardless of interference structure and regardless of which construction satisfying definition 6.2 produced the allocation. The interference-based mechanism remains available in principle for other witnesses, but it is not what the canonical witness of this monograph uses, and the sketch above is corrected accordingly.

9.4 Corollaries

Corollary 9.3 (Local Improvement Does Not Guarantee Global Sufficiency). *Improving any single stage’s local budget cannot, by*

itself, eliminate all instances of the failure mode exhibited in theorem 9.2.

Proof Sketch. Verified directly against the witness in Appendix A: increasing β_1 cannot restore a distinction already absent from A_1 by the time stage 1 acts, since proposition 6.12 forbids recovery absent root access. Only an improvement at the specific stage where the loss occurred could have prevented it, and even that requires knowing, at that stage, that the eventually-relevant task would need the distinction in question – precisely the knowledge proposition 3.7 shows is generally unavailable.

Corollary 9.4 (No Free Lunch for Relays). *A budget relay with n uncoordinated stages has cumulative discard (definition 6.11) that is non-decreasing in n (proposition 6.12), realized with equality across arbitrarily many additional pass-through stages (proposition A.8); the coordination mechanisms of Chapter 7 can narrow, but not eliminate, the resulting insufficiency (proposition 7.7).*

Proof Sketch. The non-decrease is proposition 6.12 directly, and Appendix A shows this bound is achieved with equality rather than merely respected, since pass-through stages contribute no further discard. This corollary should not be read as a claim that cumulative discard *risk* grows in some stronger, quantitative sense across relays in general; that stronger claim is not established here and is recorded as open in Appendix F. What is established is the existence half only: discard does not shrink, and can accumulate to the full extent this chapter’s witness demonstrates.

9.5 Relation to the Redistribution Theorem

theorem 9.2 is not a special case of theorem 4.6, and is not derivable from it without the budget-relay machinery of Chapter 6. The Redistribution Theorem concerns simultaneous competition for one observer’s budget; the Composition Failure

Theorem concerns a single distinction's fate across a sequence of independent, non-communicating budgets. Appendix C develops this contrast concretely rather than only structurally: pooling the witness's two stage budgets into a single shared total is shown there (proposition C.1) not merely to make theorem 4.6 applicable again, but to actually succeed at funding the distinction the separated relay was forced to drop, using the identical total nominal resource. The insufficiency this chapter's theorem exhibits is therefore a consequence specifically of definition 6.6's prohibition on transferring unspent capacity across stages, not of any absolute scarcity of resource – confirming that the two theorems occupy genuinely disjoint settings rather than one subsuming the other.

Part III

Modes of Access: Preservation and Reconstitution

Chapter 10

Preservation Mode, Reconstitution Mode, and the Fidelity Continuum

Synopsis. This chapter revisits the traditional distinction between direct access and reconstruction and replaces it with a single fidelity continuum. Every consumer interacts with some rendering, and every rendering preserves only part of the original distinction structure, so preservation becomes a limiting case of reconstitution rather than a separate ontology. Fidelity is introduced as a measure of how closely a rendering preserves an original object's admissible continuation space, and the chapter argues that reconstitution does not escape the Composition Failure Theorem: a reconstructed trace is simply a new relay stage, subject to the same constraints as any other.

10.1 The Naive Binary

An initial formulation of access modes distinguishes preservation mode – the consumer has direct causal access to the artifact itself – from reconstitution mode – the consumer has access only to traces from which structure must be regenerated. This chapter shows that the binary does not survive scrutiny and replaces it with a single continuous variable.

Remark 10.1. Nearly every case initially classified as preservation is in fact mediated by some rendering: a photocopy, a scan, a decoded audio file. A photograph of a painting is not the painting; a scan of a manuscript is not the manuscript; a decoded audio stream is not the acoustic event it encodes. Such cases are already points on the loss spectrum of definition 2.2, not a qualitatively separate category.

10.2 Fidelity, Defined

Definition 10.2 (Fidelity). For a rendering r of an object e and a task τ ,

$$F(r, \tau) = \frac{|\mathcal{A}_\tau(r) \cap \mathcal{A}_\tau(e)|}{|\mathcal{A}_\tau(e)|},$$

a measure of how closely the admissible continuation set available from r approximates the admissible continuation set available from the original object e itself, for the task τ in question. Per remark 2.1, this is stated only for the finite or countable case.

Remark 10.3 (The Intersection Simplifies). By proposition 3.2, $\mathcal{A}_\tau(r) \subseteq \mathcal{A}_\tau(e)$ always, so the intersection in definition 10.2 is simply $\mathcal{A}_\tau(r)$ itself: $F(r, \tau) = |\mathcal{A}_\tau(r)|/|\mathcal{A}_\tau(e)|$. The intersection form is retained in the definition because it makes explicit, without relying on a separately proved proposition, exactly

what is being compared; the simplification is used freely from here on.

Remark 10.4 (A Standing Assumption: Ground-Truth Sufficiency). Throughout this chapter it is assumed that $\mathcal{A}_\tau(e) \subseteq \mathcal{C}_\tau(e)$: the true object e is itself always sufficient for any task evaluated against it. This is close to definitional given how $\mathcal{C}_\tau(e)$ was fixed relative to e in definition 3.4, but it is worth stating openly, since corollary 10.6 below depends on it.

Proposition 10.5 (Preservation as a Limit Case). *“Preservation mode” is the limit of reconstitution mode as $F(r, \tau) \rightarrow 1$ for the tasks τ in question; it is not a distinct ontological category but a point on the continuum defined by definition 10.2.*

Proof Sketch. As $F(r, \tau) \rightarrow 1$, $\mathcal{A}_\tau(r)$ converges to $\mathcal{A}_\tau(e)$: the continuations available from the rendering become indistinguishable, for the purposes of τ , from those available from the original. Reconstitution from a sufficiently high-fidelity trace is then operationally indistinguishable from direct access, which is what “preservation” was intended to capture.

Corollary 10.6 (Fidelity One Implies Sufficiency). *If $F(r, \tau) = 1$, then r is sufficient for τ (definition 3.5).*

Proof Sketch. $F(r, \tau) = 1$ gives $\mathcal{A}_\tau(r) = \mathcal{A}_\tau(e)$ by remark 10.3, and $\mathcal{A}_\tau(e) \subseteq \mathcal{C}_\tau(e)$ by remark 10.4, so $\mathcal{A}_\tau(r) \subseteq \mathcal{C}_\tau(e)$, which is exactly definition 3.5.

This corollary sharpens proposition 10.5’s informal “operationally indistinguishable” into a precise claim: the fidelity continuum’s upper limit is not merely subjectively hard to tell apart from direct access, it is provably sufficient in the sense Chapter 3 already uses throughout the rest of the monograph.

Proposition 10.7 (Benign Loss Preserves Perfect Fidelity). *If every distinction in $D(r)$ is benign relative to τ (definition 2.7), then $F(r, \tau) = 1$.*

Proof Sketch. By the same argument as proposition 3.6: no distinction discarded by r is required to determine any continuation's membership in $\mathcal{A}_\tau(e)$, so no continuation reachable from e becomes unreachable from r for the purposes of τ , giving $\mathcal{A}_\tau(r) = \mathcal{A}_\tau(e)$ and hence $F(r, \tau) = 1$ by remark 10.3.

proposition 10.7 is the fidelity-continuum counterpart of proposition 3.6, and the two propositions together are why this chapter's continuous measure and Chapter 3's binary sufficiency criterion cohere rather than compete: perfect fidelity for a task and sufficiency for that task are the same claim, viewed through cardinality in one case and through set containment in the other.

10.3 The Hard Test Case: A Chess Board in Front of You

A chess board physically present to a player is the strongest available candidate for genuine, unmediated preservation. Yet perceiving the board is itself a rendering process – a visual system producing a percept from ambient light – and is therefore, by remark 10.1, already a point on the fidelity continuum rather than a case falling outside it. Whether *any* case achieves fidelity exactly equal to 1, or whether the entire category of strictly unmediated access is an idealization available only to formal or machine systems (a verifier checking a proof directly against explicit axioms, per Chapter 16) and never to embodied perception, is left open here.

Open Question 10.8. Does any instance of “direct access” to an artifact involve strictly zero mediation, or is unmediated access an idealization that applies cleanly only to formal/machine verification and never to embodied perception? Resolving this requires engaging directly with the question raised in full in Chapter 11.

10.4 Recursive Insufficiency

Proposition 10.9 (Recursive Insufficiency). *If A_i is a trace consumed in reconstitution mode, the act of reconstitution – schema plus trace producing a usable object – is itself a transition B_i in a budget relay in the sense of definition 6.1, and is therefore subject to theorem 9.2.*

Proof Sketch. Reconstitution takes a trace A_i and a schema and produces a usable object A_{i+1} for continuation. This is not merely analogous to a relay transition; it can be given the same triple structure directly. The schema’s capacity to correctly fill a given gap plays the role of C_i (some gaps are cheaply and reliably filled by strong priors, others are not); competition among candidate fillings for the same gap, or among several gaps drawing on the same limited attention or working memory, plays the role of I_i ; and whatever bounded capacity the reconstitution process has available – time, attention, cognitive resources – plays the role of β_i . Under this identification, $A_i \rightarrow A_{i+1}$ satisfies definition 6.1 exactly, not by loose analogy, and lemma A.4’s mechanism applies to reconstitution precisely as it applies to any other transition: a required feature of the reconstructed object that costs more than the reconstitution process’s available capacity is not recovered, regardless of how the schema is implemented.

This corrects a claim made informally in the research program that generated this monograph, in which reconstitution appeared briefly to “escape” the insufficiency problem by changing register from information recovery to regeneration. proposition 10.9 shows this is not the case, and shows it by identification rather than metaphor: reconstitution relocates the insufficiency problem to the trace and the schema, and theorem 9.2 applies to that relocation exactly as it applies to any other stage of a relay, because it *is* another stage of a relay.

CHAPTER 10. PRESERVATION MODE, RECONSTITUTION MODE, AND THE FIDELITY CONTINUUM

Chapter 11

Memory as a Limiting Case: MEM|8 and the Ontology of Traces

Synopsis. This chapter treats memory as both an example and a challenge to the framework. Two readings of MEM|8 are set out: a narrow reading, under which memory is simply another reconstructive process operating over traces, and a broad reading, under which memory or process is ontologically prior to representation itself, implying that all access to artifacts is already reconstructive. The chapter does not attempt to resolve this dispute. It instead analyzes the consequences of each reading and identifies which parts of the framework remain stable regardless of the outcome, treating the fork as an explicit, acknowledged dependency rather than concealing it behind a premature resolution.

This chapter is a friction chapter, not a worked example, and is written to be read as such. It does not resolve the question

it raises; it states the question precisely enough that the rest of the monograph's dependence on one answer or the other is visible, and it closes with a table stating exactly which results survive regardless of the answer and which do not.

11.1 Two Readings

MEM|8's core claim — that memory is prior to representation rather than a degraded copy of it — admits at least two readings with very different consequences for this monograph, and the difference between them is worth setting out in full before either is evaluated.

Under the narrow reading, biological recall specifically is reconstructive: a trace-based regeneration process rather than retrieval of a stored copy. This reading is well supported by the eyewitness-memory literature and is compatible with treating memory as one more point on the fidelity continuum of Chapter 10, alongside photocopies, scans, and decoded audio. On this reading, memory is unusual among renderings only in degree — its fidelity tends to be low and its schema-driven gap-filling tends to be confident about its own accuracy in a way a photocopier is not — not in kind.

Under the broad reading, representation *in general* is a stabilization of prior memory or process, not the reverse: perception, recognition, and even the sense of directly encountering an external object are themselves outputs of an ongoing memory-like process, rather than a separate, more basic capacity that memory merely draws on afterward. Under this reading, no consumer — including a chess player looking at a physical board — ever has strictly unmediated access to an external artifact, since the very act of perceiving the board is already a memory-like construction rather than a transparent window onto it. This would undermine the very idea of a fixed root object (definition 6.8) for any embodied consumer, since a root

object's usefulness (proposition 6.9) depends on there being a stable thing to re-derive from, and the broad reading denies that anything external is ever encountered other than through the same reconstructive process that governs recall.

11.2 Considerations on Each Side

It would be premature, and beyond this monograph's scope, to adjudicate between these readings, but it is worth surveying briefly what would need to be true for each to hold, so that the fork is a substantive question rather than an empty one. The narrow reading is the more conservative position and requires only what is already well established: reconstructive-memory research shows that recall is schema-driven and error-prone in characteristic, well-documented ways, without requiring any claim about perception or representation more broadly. The broad reading is a stronger, more general claim, and its most natural allies are views in cognitive science that treat perception itself as a constructive or predictive process rather than a passive registration of external stimuli — the general family of ideas sometimes described as treating perception as a controlled or constrained hallucination, or as governed by predictive processing in which incoming signal is used to correct an ongoing internally generated model rather than read off directly. If some view in this family is correct, the boundary between “perceiving the board” and “remembering the board” may not be a boundary between two different kinds of access at all, but two points on one continuous process — which is exactly what the broad reading claims and what this monograph's own Part III argued for the boundary between preservation and reconstitution more narrowly (proposition 10.5). This is a genuine structural echo, and it is tempting to treat it as evidence for the broad reading; this monograph resists that temptation, since an echo between two frameworks is not the same as a

proof that one licenses the other, and the question of whether perception and memory share a single underlying mechanism is an empirical matter this book is not positioned to settle.

11.3 Consequences of the Broad Reading

If the broad reading is correct, the admissibility and distinguishability apparatus of Part I would need to be built on top of a process-first ontology, rather than assuming a stable artifact that a consumer either does or does not have access to. This monograph explicitly declines to resolve this question here. It is stated as the single largest open dependency of the entire framework.

Open Question 11.1. Does MEM|8's claim generalize beyond biological recall (the narrow reading) to representation as such (the broad reading)? If the broad reading holds, Part I's treatment of artifacts as stable objects available for direct or indirect access requires revision from the ground up, rather than the more modest correction made in proposition 10.5.

11.4 What Survives Regardless of the Fork

Both readings agree that a trace or a percept is a finite, lossy substrate; they disagree only about whether any *non*-trace substrate exists at all for an embodied consumer. This distinction is sharp enough to sort the monograph's results cleanly into those that depend on it and those that do not.

11.5. WORKED MATERIAL UNDER THE NARROW READING

Result	Narrow reading	Broad reading
proposition 10.9	holds	holds
theorem 9.2	holds	holds
proposition 16.1 (proofs, formal case)	holds	holds
proposition 10.5	holds as stated	upper limit
open Question 10.8	resolved: no	remains open
Ch. 13's physical-board case	fidelity $\rightarrow$ 1 available	fidelity —
Embodied cases generally, Part IV	conditional	structural

The pattern in this table is not accidental. Results stated about *formal or machine* substrates — a proof checked by an algorithm against explicit axioms, a substrate chain defined over named distinctions without reference to how those distinctions come to be perceived — do not mention embodied access at all, and so cannot be disturbed by a dispute about what embodied access amounts to. theorem 9.2 in particular is stated entirely in terms of renderings, budgets, and discard sets; nothing in its proof (Appendix A) invokes a consumer's perceptual access to anything, so it is simply not the kind of claim the MEM|8 fork bears on. Results that *do* mention embodied access — most sharply, open Question 10.8 and every worked-domain discussion of a physical object sitting in front of an observer — are exactly the ones the fork can move. The table's purpose is to make that boundary explicit rather than leave it to be discovered piecemeal across later chapters.

11.5 Worked Material Under the Narrow Reading

The remainder of the monograph's treatment of memory (Chapter 19) proceeds under the narrow reading, since it is the reading supported by existing empirical literature independent of

CHAPTER 11. MEMORY AS A LIMITING CASE: MEM|8 AND THE ONTOLOGY OF TRACES

MEM|8's broader claim, and flags explicitly, wherever embodied perception is discussed in Part IV, that the broad reading remains an open alternative per open Question 11.1.

Chapter 12

Modes of Access as Task-Relative, Not Object-Relative

Synopsis. This chapter synthesizes the preceding discussion by arguing that access mode is determined by the relationship among consumer, task, and time, not by an intrinsic property of an artifact. The same object can function as a preserved source for one task while acting as an incomplete trace for another; proofs, paintings, contracts, and memories are revisited to show how a single artifact occupies different points on the fidelity continuum depending on what continuation is being attempted. This dissolves the earlier preservation/reconstitution binary into a more general relational framework.

12.1 The Criterion, Made Precise

Chapter 10 defined fidelity for a rendering r of an object e relative to a task τ . What was left implicit there is that r itself depends on who is asking and when: the substrate a particular consumer actually has in hand is not a fixed fact about the world, it is a fact about that consumer at that moment.

Definition 12.1 (Substrate Available to a Consumer). For a consumer c at time t , $r_{c,t}$ denotes the rendering (definition 2.2) c has causal access to at t .

Definition 12.2 (Access Mode). The access mode of consumer c to a task τ at time t is $M(c, \tau, t) := F(r_{c,t}, \tau)$.

Under this formalization, definition 12.2 is not an independent postulate about the nature of access; it is fidelity, already defined, simply evaluated at whichever substrate a specific consumer's specific circumstances make available. This is a stronger and more useful move than describing mode as "a relation among consumer, task, and time" in the abstract, because it says exactly what that relation *is*: not a new primitive, but an old one indexed correctly.

Proposition 12.3 (Mode as a Relation). *Access mode is a relation $M(c, \tau, t)$ over consumer c , task τ , and time t – not a property $M(A)$ of an artifact A in isolation. The same artifact can occupy different points on the fidelity continuum of definition 10.2 simultaneously for different consumer-task pairs.*

Proof Sketch. Immediate from definition 12.2: $M(c, \tau, t)$ is defined as a function of $r_{c,t}$ and τ jointly, and $r_{c,t}$ is itself consumer- and time-indexed by definition 12.1. M collapses to a property of a single artifact A alone only in the special, non-generic case where $r_{c,t} = A$ for every consumer and every time – which is exactly what fails whenever different consumers have access to different renderings of the same underlying object, or the same consumer's access changes over time.

12.2 Same Artifact, Different Modes

A formal proof is high-fidelity (definition 10.2) for the task of verification, since $r_{c,t}$ for a verifier with the text in hand is close to the proof itself; the same proof is lower-fidelity for the task of explanation or generalization, since $r_{c,t}$ for a reader attempting those tasks is typically their memory or partial recollection of the argument, not the text. A painting is, for its painter during production, close to direct access; for a later viewer inferring causal history from static cues (Chapter 14), $r_{c,t}$ is the viewer's percept of the canvas rather than the canvas itself, and how close that percept sits to the canvas is exactly the question Chapter 11's fork leaves open (open Question 11.1). A contract clause is high-fidelity for the task of reading its literal text; the same contract is low-fidelity for the task of establishing original intent, which is typically litigated by reconstructing a trace-based approximation of a past state of mind (Chapter 17) – $r_{c,t}$ for a court asking about intent is correspondence, drafting history, and testimony, not the contract text itself.

Artifact	Task	$r_{c,t}$	Fidelity
Proof	Verification	The proof text it- self	High
Proof	Explanation	Reader's recol- lection of the ar- gument	Low
Painting	Production	Canvas under the painter's hand	High
Painting	Causal infer- ence	Viewer's per- cept of the canvas	Open (open Question 11.1)
Contract	Literal reading	The contract text itself	High
Contract	Establishing in- tent	Correspon- dence, drafting history, testi- mony	Low

The table's third column is doing the real work: in every row, what changes across tasks for the same artifact is not the artifact but which substrate a consumer pursuing that particular task actually has access to, exactly as definition 12.2 formalizes.

12.3 Why This Dissolves Rather Than Answers the Earlier Question

It is worth being precise about what this chapter, and Part III as a whole, has and has not accomplished. It has not settled whether embodied perception ever achieves fidelity exactly 1; that question is left explicitly to Chapter 11's unresolved fork. What it has done is show that the original question motivating

12.3. WHY THIS DISSOLVES RATHER THAN ANSWERS THE EARLIER QUESTION

this part of the monograph — is a given artifact accessed in preservation mode or reconstitution mode? — was not well posed to begin with, because it asked for a property of the artifact when the answer was always a property of the artifact together with a consumer, a task, and a time. Once definition 12.2 makes that dependency explicit, the preservation/reconstitution distinction does not need to be defended as a real boundary somewhere on the fidelity continuum; it can simply be retired in favor of the continuum itself, with “preservation” surviving only as a convenient name for the high-fidelity end of a scale that has no other privileged points. Part IV proceeds on this basis: rather than asking of chess, painting, maps, proofs, contracts, science, memory, and media pipelines which mode of access each one exemplifies, it asks, for each domain and each task, what $r_{c,t}$ actually is and how close it sits to the source.

Part IV

Worked Domains

Chapter 13

Chess: The Closed Hidden-State Case

Synopsis. Chess serves as the canonical closed hidden-state domain. This chapter examines the gap between the visible board and the additional state required for correct continuation, such as castling rights and the repetition count, and shows that because this hidden state is finite and enumerable, augmentation restores sufficiency in principle. Chess offers a clean illustration of the framework's core concepts without the complications introduced by open-ended real-world domains, and the chapter closes by revisiting, through the case of a physical board, the question of what direct access actually means.

13.1 The Board as Substrate

A chess position's visible state – piece placement – omits castling rights, en passant eligibility, and the repetition count, each of which is required to determine legal continuations in some po-

sitions. This is the canonical witness for definition 3.8’s closed case: the omitted variables are drawn from a short, enumerable list, in sharp contrast to painting’s open causal space (Chapter 14).

Example 13.1. Two positions can have identical piece placement – the same pieces on the same squares – while differing in whether either player retains the right to castle, if one position arose from a king or rook having moved and moved back while the other did not. A rendering that records only piece placement cannot distinguish these; a rendering that also records castling rights can. This is a minimal pair in the exact sense of definition 2.4: two source states differing in one named distinction, told apart in the source, and identified by a rendering that omits it.

13.2 Augmentation Worked in Full

Augmenting the visible board with $H = \{\text{castling rights, en passant status, ...}\}$ satisfies proposition 3.9: the augmented rendering $r' = (r, H)$ is sufficient for determining every legal continuation, since H exhausts the game’s hidden state relevant to legality. This can be stated even more precisely using Appendix A’s recovery-task machinery: each element of H corresponds to a recovery task τ_h (definition A.1) – “does this side retain castling rights,” say – and lemma A.2 gives directly that r' is sufficient for τ_h if and only if $h \notin D(r')$. Since H is included in full, every such τ_h is satisfied simultaneously, which is exactly what proposition 3.9’s conclusion amounts to once stated at the level of individual recovery tasks rather than only at the level of the augmented rendering as a whole. Chess is, in this sense, the domain where the recovery-task family T_{rec} introduced to construct Appendix A’s witness happens to be finite and known in advance, rather than open – the one condition under which proposition 3.7’s impossibility result does not bind.

13.3 The Simul Player as a Degenerate Relay

A simultaneous-exhibition player moving between boards is formalized as a degenerate one-stage relay in which B_0 is deliberately reapplied fresh at each encounter with a given board, rather than carried forward across encounters: the player reads the current position cold and acts on it, with no continuity of an internal working state between visits.

Proposition 13.2 (Stateless Re-Entry Is Sufficient When the Board Is). *If the augmented rendering $r' = (r, H)$ of a chess position is sufficient for the task of determining the best legal move (proposition 3.9), then a player re-entering the game statelessly at that position – with no memory of how it arose – can determine the same best move as a player with full continuous memory of the game.*

Proof Sketch. By proposition 3.9, r' is sufficient for the task, so $\mathcal{A}_\tau(r') \subseteq \mathcal{C}_\tau(e)$ regardless of which consumer is evaluating it or what that consumer's history has been. A continuously-present player's continuation is computed from $\mathcal{A}_\tau(r')$ exactly as a freshly re-entering player's is, since neither player has access to information beyond r' once castling rights, en passant status, and repetition count are included in it. The game's history matters only insofar as it determines H 's current values, and H is, by hypothesis, already reflected in r' .

proposition 13.2 is what licenses treating the simul player as a legitimate limiting case rather than a curiosity: the entire reason walking a circle of boards and reading each one cold works at all is that chess's hidden state is closed and, when augmented, fully captured in the position itself, exactly as section 7.2's discussion of augmentation as a coordination mechanism requires. This contrasts sharply with the genuine multi-stage relays of Chapter 20, where state *is* intended to carry forward across stages that do not share a single fixed task, and where theorem 9.2's failure mode becomes possible

precisely because that carried-forward state is uncoordinated across stages with differing priorities – a condition that does not arise in the simul case because every visit to a given board shares exactly one task.

13.4 Revisiting the Hard Test Case

The physical board in front of a player is, per open Question 10.8, the strongest available candidate for fidelity $F(r, \tau) = 1$ under definition 10.2. In the vocabulary of Chapter 12, the question is precisely what $r_{c,t}$ is for a player c at the moment of moving: is it the board itself, or is it c 's visual percept of the board, already processed by whatever mechanism governs perception? If the narrow reading of chapter 11 is correct, this distinction can be drawn cleanly and $r_{c,t}$ can be treated as the board itself for practical purposes, recovering the naive intuition that a player has direct access to the game. If the broad reading is correct, $r_{c,t}$ is already a reconstructive product of the player's perceptual system, and the question of whether fidelity ever actually reaches 1 here — rather than merely approaching it closely enough for chess's purposes — is left open pending resolution of open Question 11.1. Nothing in this chapter's treatment of augmentation and stateless re-entry depends on settling that question: proposition 13.2 holds given whatever $r_{c,t}$ turns out to be, provided it includes H , and the fork bears only on how literally “the board itself” can be taken as that substrate.

Chapter 14

Painting: Open Causal Spaces and Benign Underdetermination

Synopsis. This chapter examines painting as the canonical open hidden-state domain. Paintings routinely omit vast amounts of information while remaining meaningful and useful; spatial compression, implied motion, and inferred causal history are analyzed as cases where a viewer reconstructs a rich world from sparse cues. Unlike chess, no finite augmentation can enumerate every causal history consistent with a painted configuration. The chapter argues that ambiguity here is not merely tolerated but often serves the intended task directly, making painting the clearest illustration in the monograph of the difference between corruption and benign underdetermination.

14.1 Spatial Compression

A small number of brushstrokes suggesting a cluster of leaves, and a larger arrangement of such clusters suggesting a forest, exploit a viewer's priors under definition 2.7: the omitted detail is redundant given what the visual system already supplies, not required for the viewer's task of recognizing foliage.

Remark 14.1 (Priors as Repair Cost Reduction). It is worth stating this connection to Chapter 4 explicitly rather than leaving spatial compression and repair economics as two separate vocabularies that happen to apply to the same example. A painter can afford to discard a leaf-level distinction d at negligible $C_R(d)$ precisely because the viewer's own perceptual system performs, for free and at the moment of viewing, whatever reconstruction would otherwise have been needed to supply d . The painter's low repair investment is not recklessness; it is an accurate assessment that the *effective* cost of maintaining d for this consumer's task is already near zero, since the consumer's prior does the work the painter would otherwise have had to fund.

14.2 Temporal Compression and the Bent Branch

A branch bent as if by wind is a static configuration that is low-probability under a viewer's prior model of objects at rest, forcing an inference of cause. Informally,

$$P(\text{cause} \mid \text{state}) \propto P(\text{state} \mid \text{cause}) P(\text{cause}),$$

used here descriptively, to fix intuition, rather than as a proved result of this monograph.

Proposition 14.2 (Open Causal Space). *No finite augmentation of a painting's visible configuration determines a unique causal history for that configuration.*

14.3. WHY PAINTING IS THE CANONICAL OPEN DOMAIN

Proof Sketch. For any finite augmentation set H added to the image, multiple distinct causal histories – wind, a bird’s departure, a hand’s release, accumulated then-melted snow – remain consistent with the image together with H . The causal space is therefore open in the sense of definition 3.8, and no finite H closes it.

Proposition 14.3 (Benign Ambiguity). *Multiple admissible causal histories for a painted configuration need not reduce success at the task of evocative reconstruction.*

Proof Sketch. The task of evocative reconstruction (definition 3.5 applied with $\tau = \text{“evoke a plausible dynamic scene”}$) does not require a unique history; any history in $\mathcal{A}_\tau(r)$ satisfies τ equally well. Ambiguity here is benign in the sense of definition 2.7 precisely because the task does not discriminate among the remaining admissible continuations. By proposition 10.7, this ambiguity is also compatible with $F(r, \tau) = 1$ for this task, even though the same rendering would have far lower fidelity for a task demanding the branch’s one true cause.

14.3 Why Painting Is the Canonical Open Domain

proposition 14.2 is the direct converse witness to proposition 3.10: painting demonstrates that sufficiency by augmentation is unavailable precisely where the hidden state is open, exactly as chess (chapter 13) demonstrates its availability where the hidden state is closed.

**14.4 A Preview: Closedness and Declaration
Are Independent**

It is tempting, having seen chess (closed, and its hidden state straightforwardly enumerable) and painting (open, and its causal space irreducibly ambiguous), to treat closedness and transparency as the same property. Chapter 15 shows they are not: a map depicts a domain (terrain) that is, in its full physical complexity, at least as rich as anything a painting depicts, yet a map’s discard set is closed and declared by its scale bar. The relevant two-by-two structure is worth stating here in preview, since painting and chess between them supply only two of its four cells.

	Declared discard	Undeclared discard
Closed hidden state	Maps (chapter 15)	Chess without <i>H</i>
Open hidden state	A destroyed-records archive	Painting (this chapter)

The upper-right cell is a chess position reported without its hidden state – closed, but undeclared, and hence merely incomplete rather than irreducibly ambiguous, since *H* could in principle be supplied. The lower-left cell is populated by a case not otherwise treated in this monograph: an archive that explicitly records that certain materials were destroyed, without being able to enumerate what they contained – open, since the destroyed content’s specifics are gone rather than merely omitted from a known list, but declared, since the fact and approximate scope of the loss is a matter of record. Painting occupies the remaining cell: open, and undeclared in the specific sense that a painting does not carry an explicit ledger of which causal histories it fails to distinguish, relying instead on the viewer’s tolerance for ambiguity (proposition 14.3) rather than on disclosure. Chapter 15 develops the closed-and-declared cell in full.

Chapter 15

Maps: Scale as a Declared Discard Set

Synopsis. Maps are examined here as one of the oldest and most successful examples of explicit, declared information reduction. A map’s scale functions as a declared discard set, informing the user in advance which distinctions have been intentionally omitted. The chapter shows that different scales support different continuation tasks, that insufficiency is not equivalent to error, and that a map can be perfectly adequate for one purpose while being wholly inadequate for another — underscoring how transparency alone can turn what would otherwise be corrupting loss into benign simplification.

15.1 The Scale Bar as a Prehistoric Loss Ledger

Definition 15.1 (Scale-Induced Discard). A map at scale $1:s$ discards every distinction d whose minimal pair (definition 2.4) corresponds to a real-world separation below a resolution threshold $\theta(s)$, where θ is increasing in s .

Proposition 15.2 (Scale as Declared Discard). *A map's scale declaration is an instance of the loss ledger (definition 6.13).*

Proof Sketch. By definition 15.1, a stated scale specifies, implicitly, the class of spatial distinctions below threshold $\theta(s)$ that the map does not represent. This is functionally equivalent to a declared discard set (definition 2.5) attached openly to the rendering, satisfying definition 2.7 rather than definition 2.8 for any task not requiring sub-threshold detail.

15.2 Task-Relative Sufficiency, Concretely

Example 15.3. A 1:50,000 map has $\theta(50,000)$ on the order of tens of meters: two buildings on the same block collapse to the same mark, but two distinct towns do not. This map is sufficient (definition 3.5) for route planning between towns and insufficient for locating a specific building on a specific block, without being incorrect at either task – the discarded distinctions are declared by the scale bar itself, and proposition 2.9 confirms this declared loss does not constitute corruption regardless of which task a given user brings to the map. A 1:5,000 map lowers θ by an order of magnitude, becoming sufficient for the building-level task at the cost of no longer fitting an entire province on one page – a different, but equally declared, discard set.

15.3 Completing the Taxonomy

Chapter 14 previewed a two-by-two structure crossing closed against open hidden state with declared against undeclared discard, and placed maps in the closed-and-declared cell. It is worth saying precisely why maps belong there and not in the open cell painting occupies, since terrain is, considered in the

abstract, an enormously complex physical domain – arguably as complex as anything a painting depicts.

Proposition 15.4 (Map Closedness Is a Property of the Discard Function, Not the Domain). *A map’s hidden-state space is closed (definition 3.8) regardless of the physical complexity of the terrain it depicts.*

Proof Sketch. By definition 15.1, everything a map discards is characterized by a single, known threshold $\theta(s)$ applied uniformly: the discarded distinction set is exactly $\{d : d\text{'s separation} < \theta(s)\}$, which is a well-defined, if large, set fixed by s alone. This is an enumerable characterization of what is missing, satisfying definition 3.8’s closed case, regardless of how much real-world detail falls below $\theta(s)$. Painting’s causal-history space (proposition 14.2) is open not because it is *large* but because no threshold-based or otherwise finite characterization of the missing causes exists at all; terrain’s complexity, in contrast, is fully characterized by $\theta(s)$ once the scale is fixed.

proposition 15.4 is the precise statement behind Chapter 14’s claim that closedness is a property of the *discard declaration*, not of the depicted domain’s complexity: a map’s terrain could in principle be as intricate as a painted scene, and the map would still be closed, because what makes a domain open in this monograph’s sense is the absence of any finite characterization of what is missing, not the sheer quantity of what is missing.

Chapter 16

Proofs: Sufficiency Relative to a Closed Task and System

Synopsis. This chapter addresses the strongest apparent counterexample to the monograph's thesis: a formal proof, once written, appears complete. The chapter shows this is correct precisely for verification relative to a fixed axiom system, and precisely why it stops being correct the moment explanation, intuition, generalization, pedagogy, or future theoretical change become the task instead. Proofs therefore demonstrate that task-relative sufficiency is genuinely achievable in the closed case, while simultaneously reinforcing why task boundaries matter as much as they do throughout the rest of the book.

16.1 The Strongest Apparent Counterexample

Any framework claiming that no finite substrate is generally sufficient invites the objection: but a machine-checkable proof, once written, is complete for verification. This chapter states precisely why that objection is correct as far as it goes, and precisely where it stops applying – a role this chapter is written to make explicit, since the objection would otherwise hang unanswered over the rest of the monograph.

Proposition 16.1 (Verification Sufficiency). *For a fixed axiom system S , a machine-checkable proof p is sufficient (definition 3.5) for the task of verification relative to S .*

Proof Sketch. The verification algorithm for S is finite and decidable by construction; every distinction it requires to certify p appears explicitly in p together with S 's axioms. No distinction required by the verification task is discarded, so definition 3.5 is satisfied.

Corollary 16.2 (Explanation Is Not Guaranteed). *proposition 16.1 does not extend to the tasks of explanation, generalization, or adaptation to a changed axiom system.*

Proof Sketch. These tasks lie outside the closed task space $\{\tau_{\text{verify}}\}$ for which S and p were held fixed in proposition 16.1's hypothesis; by proposition 3.7, sufficiency for tasks outside a closed, anticipated set is not guaranteed by a finite substrate.

16.2 Why This Resolves Rather Than Merely Sidesteps the Objection

proposition 16.1 shows the impossibility claim of principle 1.1 was never intended to apply where both the task and the formal

system are closed and fixed in advance; corollary 16.2 shows that the moment either is allowed to vary, the same proof reverts to an ordinary instance of proposition 3.7. Sufficiency, throughout this monograph, is always sufficiency relative to a declared task space, and a proof's celebrated reliability is precisely a case where that task space has been closed by convention.

16.3 Why Proofs Do Not Exhibit Composition Failure

Chapter 6 onward has treated failure as arising when a chain's stages pursue *different* local tasks $\tau_0, \dots, \tau_{n-1}$, each locally reasonable, that jointly fail to anticipate a later τ^* . A proof's transmission – through typesetting, copying, publication, translation of notation – is also a chain of renderings in the sense of definition 6.1. It is worth stating precisely why this particular chain typically avoids theorem 9.2's failure mode for the verification task, even though it is exactly the kind of chain the theorem is about in general.

Proposition 16.3 (Homogeneous-Task Chains Preserve What They Share). *If every stage of a substrate chain shares the same task, $\tau_0 = \tau_1 = \dots = \tau_{n-1} = \tau$, and every stage's budget satisfies $\beta_i \geq R_\tau$ (the total cost of τ 's required distinctions), then A_n is sufficient for τ .*

Proof Sketch. By definition 9.1, each stage ranks τ 's required distinctions above all others, and by hypothesis $\beta_i \geq R_\tau$ at every stage, so every required distinction is funded at every stage: no required distinction is ever dropped, by definition 6.2. Since this holds for every i , τ 's required distinctions survive intact into A_n , and A_n is sufficient for τ by definition 3.5.

proposition 16.3's hypothesis is exactly what typesetting and transmission processes for mathematical proofs typically

satisfy for $\tau = \tau_{\text{verify}}$: every party handling the text – a typesetter, a journal, a translator of notation – treats preserving the logical content as the overriding priority, well-resourced enough that $\beta_i \geq R_{\tau_{\text{verify}}}$ at every stage. This is why proofs do not feel like an exception requiring special pleading: proposition 16.3 shows that a chain avoids composition failure for a given task exactly when every stage treats that task as priority throughout, which is the normal condition for a proof’s transmission and verification, but conspicuously is *not* the normal condition for a proof’s transmission with respect to explanation. Nothing about the transmission chain prioritizes preserving the motivating intuition, the historical context of the discovery, or the informal picture that made the result plausible in the first place; those distinctions are exactly the ones a typesetting or publication process has no particular reason to fund, and corollary 16.2 is the consequence.

16.4 Verification as Preservation, Explanation as Reconstitution

A proof text is high-fidelity (definition 10.2) for verification, since $r_{c,t}$ for a verifier with the text in hand is close to the proof itself (definition 12.2); understanding *why* the proof works is typically reconstitution-mode, built from a reader’s trace-based reconstruction of the argument’s structure, and is therefore subject to proposition 10.9 in exactly the way memory is (Chapter 19).

Chapter 17

Legal Contracts: Adjudication as Non-Informational Resolution

Synopsis. Contracts attempt to regulate future contingencies through finite language, making them a natural example of continuation under uncertainty. This chapter examines how legal systems respond when a contract encounters a situation its authors did not anticipate, and presents adjudication as a distinctive mechanism: a court resolves ambiguity through authoritative decision rather than by recovering additional distinctions about the past. Law is, on this framework, a domain in which continuation proceeds despite persistent, unresolved informational insufficiency.

17.1 Contracts as Attempted Brute-Force Closure

A contract's enumerated contingencies ("in the event of X , Y , or Z ") attempt to close what is, in fact, an open contingency space by brute enumeration, resembling augmentation (section 7.2) in strategy while targeting a domain that does not satisfy definition 3.8's closed case.

Example 17.1. A force majeure clause enumerating war, natural disaster, and government action as excusing conditions is an augmentation set $H_{\text{contract}} = \{\text{war, natural disaster, government action}\}$ in exactly the sense of proposition 3.9: it pre-commits the parties' agreement to cover these named contingencies without needing to anticipate, at the time of drafting, which one might occur. The clause is sufficient for any dispute falling within H_{contract} . A disruption outside this list – one neither party had reason to name at drafting time – is not covered, not because the clause was drafted carelessly, but because H_{contract} , like any finite augmentation, cannot exhaust an open space of future circumstances.

Proposition 17.2 (Contractual Augmentation Targets the Wrong Kind of Domain). *A contract's enumerated-contingency strategy achieves proposition 3.9's sufficiency only for contingencies within its enumerated set H_{contract} ; by proposition 3.10, no finite H_{contract} achieves sufficiency for the full space of future circumstances, since that space is open in the sense of definition 3.8.*

Proof Sketch. Immediate from proposition 3.10 applied with the future-circumstance space playing the role of the open hidden-state space: augmentation is the correct strategy in form, exactly as it is for chess (chapter 13), but the domain it is applied to lacks the closure property chess's hidden state has, so the strategy's guarantee does not transfer.

This mismatch between strategy and domain is, on the framework developed here, a structural explanation for the persistence of contractual litigation: no finite enumeration of contingencies exhausts an open space of future circumstances, and every contract that enumerates contingencies is, in this precise sense, applying chess's solution to painting's problem.

17.2 Adjudication, Worked

proposition 7.5 is developed here in full: a court's ruling on a disputed clause selects a binding continuation without recovering any additional distinguishability information about the parties' original intent. The information deficit – what the parties actually meant at the time of drafting – remains after the ruling exactly as before it; what has changed is that further continuation is now permitted to proceed on the ruling's basis. In the vocabulary of Section 7.5, the court substitutes its ruling as the operative correctness set $\mathcal{C}_\tau(e)$ going forward, regardless of whether that ruling coincides with whatever the undiscoverable true correctness set would have contained.

17.3 Litigating Intent as Reconstitution

Establishing original intent is a reconstitution-mode task (chapter 12): for a court asking about intent, $r_{c,t}$ is correspondence, drafting history, and testimony, none of which constitutes direct access to the intent itself (definition 12.1). This is a further instance of proposition 10.9: the traces used to reconstruct intent are themselves finite substrates subject to theorem 9.2, which is part of why intent disputes resist final resolution by evidence alone and are ultimately settled, per proposition 7.5, by authority rather than by information recovery.

CHAPTER 17. LEGAL CONTRACTS: ADJUDICATION AS NON-INFORMATIONAL RESOLUTION

Chapter 18

Scientific Theories: Revision as Relay Replacement

Synopsis. Scientific theories are treated here as finite renderings of an open empirical world. The chapter argues that a theory inevitably encounters observations it did not anticipate and therefore cannot remain sufficient indefinitely; scientific progress is reframed as relay replacement rather than local correction, with a new theory replacing an old relay once accumulated insufficiency becomes too great to ignore. Unlike most domains treated in the monograph, science actively depends on discovering its own insufficiency, making it a particularly revealing test of the framework.

18.1 A Theory as a Finite Substrate for an Open Observation Space

Proposition 18.1 (Theories Are Renderings of an Open Observation Space). *A scientific theory is a finite rendering of a potentially unbounded space of future observations.*

Proof Sketch. The observation space relevant to a theory is not fixed at the theory's formulation; new instruments, new regimes, and new experimental designs continually generate previously unavailable observations. A theory, being a finite statement, cannot represent this space exhaustively, and definition 2.2's projection structure therefore applies.

Proposition 18.2 (Revision Necessity). *Every finite theory admits possible future observations that would require its revision.*

Proof Sketch. This is proposition 3.7 applied to proposition 18.1's open observation space: a future observation τ^* can always be constructed that the theory, as a finite rendering, was not built to accommodate. This is essentially the falsifiability criterion, reframed within the admissibility apparatus of Chapter 3 rather than derived independently.

18.2 Revision, Not Augmentation

Unlike chess (chapter 13), where insufficiency is resolved by augmenting a closed hidden-state set, and unlike maps (chapter 15), where insufficiency is resolved by declaring the discard, a theory's response to proposition 18.2 is typically wholesale replacement of the theory itself (definition 7.6, section 7.6), not a patch to the existing substrate. This is the one domain treated in this monograph where global insufficiency, per principle 1.1, is not merely tolerated but functions as the intended mechanism of progress: a theory's eventual insufficiency is what licenses its replacement by a better one.

Example 18.3. A theory calibrated against observations within a certain regime – low relative velocities, say – can be sufficient (definition 3.5) for every task posed within that regime while being straightforwardly insufficient for tasks posed in a regime its observational basis never covered. The historical replacement of such a theory by one validated across a wider observational range is a canonical instance of definition 7.6: the newer theory is not derived from the older one by augmentation, in the sense proposition 3.9 would require, since the older theory’s observational basis was never closed to begin with; it is a new production built from an enlarged evidential base.

18.3 A Note on Adjacent Debates

This chapter’s claim is narrower than it might appear next to some well-known positions in philosophy of science, and it is worth marking the boundary rather than letting the two blur together. proposition 18.2 claims only that a finite theory cannot anticipate every future observation, which is a direct instance of proposition 3.7 and carries no commitment about whether successive theories are commensurable, whether observation itself is theory-laden in ways that complicate what counts as a “new” observation, or whether scientific change is better described by gradual accumulation or by more discontinuous conceptual shifts. These are substantive, contested questions in their own right, and this monograph does not adjudicate them; definition 7.6’s formal claim – that a superseded chain is replaced rather than repaired – is compatible with more than one position on how sharp or continuous that replacement turns out to be.

Chapter 19

Memory: Reconstitution and the Recursive Limit

Synopsis. This chapter returns to memory as a concrete illustration of reconstitution. Memories are analyzed as traces requiring reconstruction through schemas, priors, and contextual cues; eyewitness testimony and autobiographical recall are examined as continuation under severe informational constraint. The chapter shows that memory does not escape the framework's central limitations but reproduces them in concentrated form, making visible, at the scale of a single mind, dynamics that are otherwise easiest to see in large institutional relay systems.

This chapter proceeds under the narrow reading of MEM|8 identified in Chapter 11; wherever a claim below would be affected by the broad reading instead, this is noted explicitly rather than left implicit.

19.1 Memory as a Three-Stage Relay

Standard treatments of memory in cognitive psychology distinguish encoding, consolidation, and retrieval as separate processes, each with its own resource constraints. This structure maps directly onto definition 6.1.

Construction 19.1 (Memory as a Substrate Chain). Let A_0 be an experienced event, A_1 the encoded trace immediately following it, A_2 the consolidated trace some time later, and A_3 the recollection produced at retrieval. The transition $A_0 \rightarrow A_1$ is governed by encoding resources (attention, salience) as its budget β_0 ; $A_1 \rightarrow A_2$ by consolidation resources (rehearsal, sleep, interference from subsequent events) as β_1 ; and $A_2 \rightarrow A_3$ by retrieval resources (available cues, time pressure) as β_2 .

Each transition in construction 19.1 satisfies definition 6.6's independence condition in the ordinary case: an event's encoding is not typically decided with reference to what a much later retrieval attempt will need, consolidation proceeds according to its own resource constraints without reference to encoding's particular priorities, and a specific retrieval attempt draws on whatever cues are available at that moment without renegotiating what consolidation preserved. construction 19.1 therefore is not merely analogous to a budget relay, in the way Chapter 10 was careful to establish for reconstitution generally (proposition 10.9): it is a three-stage instance of one, with each stage's budget independently and plausibly set.

Proposition 19.2 (Memory Retrieval Is a Relay Transition). *The transition trace $\rightarrow$ recollection satisfies the conditions of a substrate-chain transition (definition 6.1).*

Proof Sketch. Retrieval takes a residual trace and a current schema and produces a usable recollection, incurring costs (effort, plausibility constraints supplied by the schema) and

losses (aspects of the original event the trace does not constrain) in the sense of definition 6.2. This satisfies definition 6.1 directly, and is the final stage of construction 19.1.

19.2 Schema-Driven Confident Error

Eyewitness-memory research documents confident, richly reconstructed testimony that is nonetheless factually wrong: the schema fills a gap the trace does not constrain, and the result is presented, to the rememberer and to others, as though it were retrieved rather than regenerated. This is a concrete instance of proposition 10.9’s “priors patch” (section 7.4) operating at retrieval time on an already-insufficient trace, and is structurally the same failure mode as proposition 14.2’s painting case, transposed from an external image to an internal reconstruction. In terms of construction 19.1, this is exactly the pattern theorem 9.2 predicts: a distinction lost or degraded at encoding or consolidation – because τ_0 or τ_1 , whatever was salient or well-rehearsed at the time, did not prioritize it – is unavailable to retrieval regardless of how confidently retrieval’s own schema fills the resulting gap, since confidence at A_3 cannot restore what A_1 or A_2 never carried forward.

Proposition 19.3 (Memory Does Not Escape Composition Failure). *Memory’s reconstitutive character does not constitute an exception to theorem 9.2.*

Proof Sketch. By proposition 19.2 and construction 19.1, memory retrieval is a relay transition, indeed the final stage of a three-stage relay; by proposition 10.9, this transition inherits the open-task-space vulnerability of any other stage in a budget relay. theorem 9.2 therefore applies to memory exactly as it applies to the media-pipeline case of Chapter 20, contrary to an informal suggestion, made earlier in the research program that produced this monograph, that reconstitution somehow

CHAPTER 19. MEMORY: RECONSTITUTION AND THE RECURSIVE LIMIT

“changes the game” rather than relocating the same problem to a new substrate.

Chapter 20

Media Pipelines: Budget Relays Made Concrete

Synopsis. This chapter applies the full framework to a real transformation chain: audio recording, signal processing, transcription, timestamp generation, summarization, and archiving, each analyzed as a stage in a budget relay preserving some distinctions while discarding others according to local priorities. The chapter provides a concrete, constructive witness for the Composition Failure Theorem by exhibiting a distinction lost early in the chain that becomes critical for a later task no stage anticipated, and closes with practical recommendations for provenance design, root-object preservation, and re-derivable archival systems.

20.1 The Full Worked Chain, Named and Numbered

The chain $\text{audio} \rightarrow \text{transcript} \rightarrow \text{summary} \rightarrow \text{archive}$ is mapped directly onto definition 6.1’s notation, but rather than leave the distinctions involved abstract, as Appendix A’s canonical witness does, this section fixes concrete, named ones, so that the constructive witness promised by this chapter’s synopsis is worked with real content rather than only with placeholders standing in for it.

Construction 20.1 (Named Pipeline Instance). Let A_0 carry four named distinctions from a recorded utterance: lexical content d_{lex} (cost 3), timestamp precision d_{time} (cost 1), sentence-level structure d_{struct} (cost 2), and emotional tone d_{tone} (cost 4). The transcription stage has task τ_0 requiring exactly $\{d_{\text{lex}}, d_{\text{time}}\}$, with budget $\beta_0 = 4$. The summarization stage has task τ_1 requiring exactly $\{d_{\text{lex}}\}$, with budget $\beta_1 = 3$.

Proposition 20.2 (The Named Instance Exhibits Composition Failure). *In construction 20.1, d_{tone} is dropped at stage 0 under any construction satisfying definition 6.2, and the resulting archive is insufficient for any task recovering d_{tone} .*

Proof. The required set at stage 0, $\{d_{\text{lex}}, d_{\text{time}}\}$, has cost $R_0 = 3 + 1 = 4 = \beta_0$ exactly. The non-required distinctions have costs $C(d_{\text{struct}}) = 2 > \beta_0 - R_0 = 0$ and $C(d_{\text{tone}}) = 4 > 0$; by lemma A.4, both are dropped under any construction, so $D_0 = \{d_{\text{struct}}, d_{\text{tone}}\}$. At stage 1, A_1 retains only $\{d_{\text{lex}}, d_{\text{time}}\}$; the required set $\{d_{\text{lex}}\}$ has cost $R_1 = 3 = \beta_1$ exactly, and the sole non-required distinction present, d_{time} , has cost $1 > \beta_1 - R_1 = 0$, so $D_1 = \{d_{\text{time}}\}$ by the same lemma. A_2 retains only $\{d_{\text{lex}}\}$. Let $\tau^* = \tau_{d_{\text{tone}}}$, the recovery task (definition A.1) for whether the speaker’s tone on a given claim was, for instance, sincere or sarcastic. Since $d_{\text{tone}} \in D_0 \subseteq D(A_0 \Rightarrow A_2)$, by lemma A.2 A_2 is insufficient for τ^* . Neither τ_0 nor τ_1 is τ^* : both are preservation

tasks over different, disjoint required sets not including d_{tone} 's recovery. $\square$

This is Appendix A's abstract construction re-run with distinctions a reader can recognize without translation: a transcription system tuned for lexical accuracy and rough timing, and a summarizer tuned for gist, jointly and irreversibly discard whether a speaker meant something sincerely or sarcastically – not through negligence at either stage, but because neither stage's task ever prioritized it, exactly as theorem 9.2 predicts.

20.2 Fan-Out, Concretized

construction 20.1 extends directly to the fan-out case of Chapter 8 by adding a second branch from A_0 : an acoustic-tone-analysis branch Y with task τ_Y requiring exactly $\{d_{\text{tone}}\}$, budget $\beta_Y = 4$ (funding d_{tone} exactly), dropping everything else. This is construction 8.4 instantiated with named distinctions: branch X (transcription, as above) and branch Y (tone analysis) are both locally admissible, both derived from the identical A_0 , and disagree silently in precisely definition 8.3's sense – a consumer relying on the transcript alone has no signal that tone was ever at issue, while branch Y , unconsulted, would have flagged it directly.

Proposition 20.3 (Pipeline Witness). *There exists an explicit audio $\rightarrow$ transcript $\rightarrow$ summary chain and a task τ^* , constructible without appeal to hypothetical or idealized stages, such that τ^* requires a distinction discarded upstream and no downstream stage's loss ledger records enough about that distinction for a consumer pursuing τ^* to detect the gap.*

Proof Sketch. construction 20.1 together with proposition 20.2 constitutes exactly this witness, with $\tau^* = \tau_{d_{\text{tone}}}$ constructed explicitly rather than posited. This result was labeled a Theorem

in an earlier draft; the label has been corrected to Proposition, consistent with the monograph’s epistemic-labeling discipline, since it is a single explicit instantiation of the general result already proved in Appendix A rather than an independent derivation.

20.3 Practical Recommendations

Preserving the raw audio as a root object (definition 6.8) and treating every downstream rendering as disposable and re-derivable directly operationalizes proposition 6.9: a discovered insufficiency at the summary or transcript stage can then be repaired by re-deriving a new rendering from the root under a revised discard policy, rather than requiring recovery of information that no longer exists anywhere in the chain. In terms of construction 20.1, this means that discovering $\tau_{d_{\text{tone}}}$ ’s relevance after the fact is recoverable, at the cost of reprocessing, if and only if the raw audio was retained; it is permanently unrecoverable if only the transcript or summary survived.

A budget-aware loss ledger (definition 6.13) attached to each stage should record not merely that d_{struct} and d_{tone} were dropped at transcription, but the budget $\beta_0 = 4$ and required set $\{d_{\text{lex}}, d_{\text{time}}\}$ that produced the drop, so a later archivist can distinguish a distinction dropped because it was judged irrelevant from one dropped purely because the transcription budget was small – exactly the distinction definition 6.13 was built to preserve. Where a fan-out structure exists, as in Section 7.3’s sense of provenance-sharing extended across branches rather than only along a single chain, cross-referencing each branch’s ledger against the others makes definition 8.3’s failure mode detectable after the fact, even though, as proposition 7.4 showed, detectability alone does not prevent the original loss.

Part V

Synthesis

Chapter 21

The Closing Statement

Synopsis. This closing chapter integrates the monograph's concepts into a single narrative: distinguishability explains what survives; admissibility explains what remains possible; repair economics explains why a distinction was or was not maintained; budget relays explain how losses accumulate across time and institutions; the fidelity continuum unifies preservation and reconstitution; and Composition Failure explains why locally rational decisions can still produce globally insufficient outcomes. The chapter closes by naming what remains unresolved, above all the MEM|8 dependency, and restates the central claim that continuation is constrained not by malice or error but by the unavoidable limits of finite repair budgets operating in an open future.

21.1 Restating the Arc

Distinguishability (chapter 2) established what survives a projection. Admissibility (chapter 3) established what continuations remain reachable given what survives. Repair economics

(chapter 4) established why a given distinction was or was not maintained, and, via corollary 4.7, established that this monograph could not simply inherit its central claim from *Conservation of Ambiguity*. The budget relay (chapter 6) formalized how discard compounds across uncoordinated stages, and the Composition Failure Theorem (theorem 9.2) proved the diachronic analogue the Redistribution Theorem does not cover. The fidelity continuum (chapter 10) collapsed preservation and reconstitution into a single variable, conditional on the unresolved question raised in Chapter 11.

21.2 The Closing Statement

Conjecture 21.1 (Closing Statement). Continuation depends on the substrate available to a consumer, whether preserved with high fidelity or regenerated from traces at low fidelity; in either case the substrate is bounded by the distinctions its production budget could afford to maintain. Because relays compose without guaranteed coordination (definition 6.6) and because future task spaces remain open (proposition 3.7), no finite chain of substrates can be guaranteed sufficient for arbitrary future continuation – not because information is destroyed, but because no stage can afford, or is positioned, to preserve everything a task it cannot yet foresee will require.

This statement is labeled a conjecture rather than a theorem deliberately: it synthesizes theorem 9.2 together with the fidelity continuum of Chapter 10, but the synthesis itself, as a single unified statement, has not been given an independent proof beyond what its components already establish.

21.3 What Remains Open

Appendix F consolidates every conjecture and open question raised in the body. The largest is open Question 11.1, on which the scope of Part III's results depends. The tightness of corollary 9.4's growth bound is conjectural. Whether a general coordination protocol across budget relays could bound, though not eliminate, theorem 9.2's failure mode is left as a direction for future work rather than claimed here.

21.4 What This Monograph Does Not Claim

This monograph does not claim that all cognition is reconstructive; it restricts that claim, via chapter 11's narrow reading, to cases independently supported by existing literature. It does not claim that *Conservation of Ambiguity's* Redistribution Theorem is superseded; theorem 4.6 is imported and used, not revised. It does not claim to have resolved the preservation/ reconstitution question definitively, since that resolution remains contingent on open Question 11.1.

Appendix A

Full Proof of the Composition Failure Theorem

Synopsis. This appendix gives a complete proof of Theorem 9.2, together with the explicit construction on which it rests. The proof is built to satisfy the robustness requirement raised in remark 6.5: rather than exhibiting a witness that happens to work under one convenient local discard rule, lemma A.4 below shows that the witnessing relay’s key discard is forced under *any* construction obeying definition 6.2, closing that specific open question for the purposes of this theorem even though the broader question of whether every claim in this monograph enjoys the same robustness remains open (appendix F).

A.1 Recovery Tasks and the Witnessing Task Family

The theorem requires an explicit, non-enumerable-in-advance family of tasks T , together with a specific $\tau^* \in T$ for which insufficiency can be verified directly rather than merely asserted. Both are supplied by a single, simple family of tasks built directly from the minimal-pair machinery of Chapter 2.

Definition A.1 (Recovery Task). For a named distinction d with minimal pair $\{x_d, x_{\neg d}\}$ (definition 2.4), the recovery task τ_d is the task of correctly reporting, given a substrate, whether the source object e satisfies x_d or $x_{\neg d}$. Its correctness set (definition 3.4) is the singleton containing whichever of the two reports matches e 's actual state.

Lemma A.2 (Sufficiency for a Recovery Task Iff No Discard). *A rendering r is sufficient (definition 3.5) for τ_d if and only if $d \notin D(r)$.*

Proof. If $d \notin D(r)$, the pair $\{x_d, x_{\neg d}\}$ is not identified by r (definition 2.3, definition 2.5), so r distinguishes which of the two states e occupies, and the correct report is the unique continuation reachable: $\mathcal{A}_{\tau_d}(r) = \mathcal{C}_{\tau_d}(e)$, satisfying definition 3.5. If $d \in D(r)$, the pair is identified, so a consumer with access only to r cannot rule out either report: both the correct and the incorrect report are reachable, $\mathcal{A}_{\tau_d}(r) \supsetneq \mathcal{C}_{\tau_d}(e)$, and definition 3.5's containment fails. $\square$

Definition A.3 (Witnessing Task Family). $T_{\text{rec}} := \{\tau_d : d \text{ ranges over the open distinctions}\}$

T_{rec} is open in the sense required by proposition 3.7: the argument given there — that a task requiring a distinction not represented in a fixed candidate list can always be constructed after the fact — applies verbatim to T_{rec} , since the space of possible minimal pairs (definition 2.4) that might later be named as relevant is not fixed in advance by any finite specification of a substrate chain's producers.

A.2 A Construction-Independent Exhaustion Lemma

remark 6.5 showed that definition 6.2 is satisfied by more than one non-equivalent construction. The following lemma isolates a sufficient condition for a discard to be forced under every such construction, so that the theorem's witness does not depend on which one a given stage happens to use.

Lemma A.4 (Exhaustion Forces Discard). *Fix a stage with budget β_i and suppose the distinctions required by τ_i (in the sense of definition 2.7) have total cost $R_i = \sum_{d \text{ required}} C_i(d) \leq \beta_i$. If every non-required distinction d' satisfies $C_i(d') > \beta_i - R_i$, then every non-required distinction is dropped – $d' \in D_i$ for every such d' – under any construction satisfying definition 6.2 together with the requirement that required distinctions are ranked strictly above non-required ones.*

Proof. Any construction satisfying definition 6.2 must respect the budget constraint $\sum_{d \in S} C_i(d) \leq \beta_i$ for the funded set S , and, by hypothesis, must rank every required distinction above every non-required one. Since the required distinctions have total cost $R_i \leq \beta_i$, funding all of them is feasible and, given their priority, is favored by both construction 6.3 (which processes higher-priority items first and encounters no obstruction, since $R_i \leq \beta_i$) and construction 6.4 (for which funding every required item dominates any allocation that sacrifices one, provided required items are assigned value strictly exceeding any combination of non-required items, which the priority requirement above entails). After funding the required set, the remaining budget is exactly $\beta_i - R_i$. By hypothesis, $C_i(d') > \beta_i - R_i$ for every non-required d' , so no non-required distinction can be funded in addition to the required set under either construction, nor under any construction obeying the same budget constraint, since funding any single such d'

alongside the required set would require total expenditure exceeding β_i . Hence $d' \in D_i$ for every non-required d' , regardless of which construction produced the allocation. $\square$

lemma A.4 is the appendix's answer to the robustness question raised in Chapter 6: it does not show that every budget relay behaves identically under every construction, but it identifies a class of stages – those where the required set exactly, or near-exactly, exhausts the budget – whose discard behavior is construction-independent, and the witness built below is chosen to fall into exactly this class.

A.3 Recalling Local Admissibility

Chapter 9 makes precise, in definition 9.1, what Theorem 9.2 means by a transition being “locally admissible with respect to its own stage’s task τ_i .” That definition does not privilege construction 6.3 over construction 6.4, or vice versa; it constrains only the priority ordering’s relationship to τ_i ’s own stated requirements, consistent with remark 6.5’s observation that definition 6.2 constrains outcomes, not procedures. It is used throughout the remainder of this appendix exactly as stated there.

A.4 The Canonical Witness

Construction A.5 (Two-Stage Witness). Let A_0 carry five distinctions $d_1, \dots, d_5$ with costs $C(d_1) = 1$, $C(d_2) = 2$, $C(d_3) = 3$, $C(d_4) = 2$, $C(d_5) = 4$ – the same instance introduced in section 6.8 and section 4.5. Let τ_0 require exactly $\{d_1, d_2\}$, with $\beta_0 = 4$. Let τ_1 require exactly $\{d_1, d_2\}$ as well, with $\beta_1 = 3$.

Proposition A.6 (The Witness Is Locally Admissible and Forces the Same Discard Under Any Construction). *In construction A.5:*

(a) B_0 and B_1 are locally admissible (definition 9.1) with respect to τ_0 and τ_1 respectively, under any construction satisfying definition 6.2; and (b) $D_0 = \{d_3, d_4, d_5\}$ and $D_1 = \emptyset$ regardless of which such construction is used.

Proof. For stage 0: the required set $\{d_1, d_2\}$ has total cost $R_0 = 1 + 2 = 3 \leq \beta_0 = 4$. Every non-required distinction has cost exceeding $\beta_0 - R_0 = 1$: $C(d_3) = 3 > 1$, $C(d_4) = 2 > 1$, $C(d_5) = 4 > 1$. lemma A.4 applies directly, giving $D_0 = \{d_3, d_4, d_5\}$ under any construction satisfying definition 6.2 with required distinctions ranked above non-required ones – which is exactly definition 9.1’s condition, so (a) and (b) both hold for stage 0.

For stage 1: A_1 retains only $\{d_1, d_2\}$ (nothing else survived stage 0), both required by τ_1 , with total cost $R_1 = 3 = \beta_1$ exactly. There are no non-required distinctions present at this stage to consider, so the exhaustion hypothesis of lemma A.4 is vacuously about an empty set, and any construction funds $\{d_1, d_2\}$ in full (the only feasible allocation using the entire budget), giving $D_1 = \emptyset$ regardless of construction. Local admissibility for stage 1 follows immediately, since there is nothing non-required present to rank incorrectly. $\square$

A.5 Extension to Arbitrary Relay Length

Construction A.7 (Pass-Through Stage). A stage j is a pass-through stage if τ_j requires exactly the distinctions present in A_j , and β_j equals their total cost exactly.

Proposition A.8 (Extension to Arbitrary Length). *For every $n \geq 2$, there is an n -stage budget relay, every transition of which is locally admissible, with $D(A_0 \Rightarrow A_n) = \{d_3, d_4, d_5\}$ exactly as in construction A.5.*

Proof. Take construction A.5’s two stages as stages 0 and 1, and insert $n - 2$ pass-through stages (construction A.7) at positions

$2, \dots, n - 1$, each requiring exactly $\{d_1, d_2\}$ at cost 3, matching A_1 's (and every subsequent stage's) content exactly. Each pass-through stage is locally admissible by the same vacuous argument given for stage 1 in the proof of proposition A.6, and discards nothing, so $D_j = \emptyset$ for $j \geq 2$. By definition 6.11, $D(A_0 \Rightarrow A_n) = D_0 \cup D_1 \cup \dots \cup D_{n-1} = \{d_3, d_4, d_5\} \cup \emptyset \cup \dots \cup \emptyset = \{d_3, d_4, d_5\}$. $\square$

A.6 Proof of Theorem 9.2

Proof. Fix any $n \geq 2$ and take the n -stage relay of proposition A.8, with every transition locally admissible by construction. Let $\tau^* := \tau_{d_5} \in T_{\text{rec}}$ (definition A.1, definition A.3). By proposition A.6 and proposition A.8, $d_5 \in D(A_0 \Rightarrow A_n)$, so $d_5 \in D(A_n)$ relative to A_0 . By lemma A.2, A_n is insufficient for τ_{d_5} .

It remains to check $\tau^* \notin \{\tau_0, \dots, \tau_{n-1}\}$. Every τ_i in the construction is a preservation task, requiring exactly $\{d_1, d_2\}$, with correctness set consisting of continuations that retain both; $\tau^* = \tau_{d_5}$ is a recovery task in the sense of definition A.1, with correctness set consisting of a single report about d_5 alone. These are distinct formal objects with distinct correctness sets for any nontrivial choice of underlying object e , so $\tau^* \neq \tau_i$ for any i .

T_{rec} is an open task space containing τ^* , as argued following definition A.3. This exhibits, for every $n \geq 2$, a budget relay and an open task space $T = T_{\text{rec}}$ satisfying every clause of Theorem 9.2. $\square$

A.7 Verifying the Corollaries

Verification of corollary 9.3. Suppose β_1 is increased in the witness of construction A.5, holding $\beta_0 = 4$ fixed. Stage 1 receives

only $A_1 = \{d_1, d_2\}$ regardless of β_1 's value, since d_3, d_4, d_5 were already absent from A_1 by the time stage 1 acts; no increase in β_1 can fund a distinction that is not present to be funded. $D(A_0 \Rightarrow A_n)$ is therefore unaffected by any change to β_1 (or to any β_j for $j \geq 1$), confirming that only an improvement to β_0 specifically – the stage where the loss actually occurred – could have prevented the eventual insufficiency, and even that requires τ_0 's producer to have known, at stage 0, that d_5 would later matter: precisely the knowledge proposition 3.7 shows is unavailable in general. $\square$

Partial verification of corollary 9.4. The existence half of corollary 9.4 – that cumulative discard is non-decreasing in relay length – is exactly proposition 6.12, and proposition A.8 shows this bound is not merely non-strict but can be realized with equality across arbitrarily many additional stages (the pass-through stages contribute no further discard). This appendix does not, however, establish the stronger, quantitative claim that cumulative discard *risk* grows across relays in general, only that it does not shrink in this particular family of constructions. The tightness question — whether a sharper growth bound holds under specific coordination mechanisms – remains open, exactly as flagged in Appendix F, and is not resolved here. $\square$

A.8 What This Proof Does Not Establish

This appendix proves Theorem 9.2 and corollary 9.3 in full, and establishes the existence half of corollary 9.4. It does not establish that every budget relay of practical interest falls into the construction-independent class identified by lemma A.4; relays in which the required set does not closely exhaust the local budget may genuinely behave differently under construction 6.3 than under construction 6.4, and nothing here rules that out for such cases. It also does not establish corollary 9.4's

APPENDIX A. FULL PROOF OF THE COMPOSITION FAILURE THEOREM

quantitative tightness, address the infinite-relay case flagged in Chapter 6, or extend to the branching relays of Chapter 8, whose fan-out structure introduces the further complication of proposition 8.6 and is not treated by the linear construction given here. Appendix F is updated to reflect exactly this scope.

Appendix B

Notation Reference

Synopsis. This appendix collects the symbols, definitions, and formal objects used throughout the monograph into a single reference table.

Symbol	Meaning
$(X, \sim_X)$	Distinction space with equivalence structure (Ch. 2)
r, A_i	A rendering / stage in a substrate chain
$D(r)$	Discarded distinction set of rendering r (definition 2.5)
$\mathcal{A}_\tau(r)$	Admissible continuation set for task τ given r (definition 4.1)
$C_R(d)$	Repair cost of distinction d (definition 4.2)
$A(d)$	Operational ambiguity of d (definition 4.3)
$I(d_i, d_j)$	Repair interference between d_i and d_j (definition 4.5)
Φ	Repair budget functional (theorem 4.6)
$B_i = (C_i, I_i, \beta_i)$	Local budget triple at relay stage i (definition 6.1)
$L_i = (D_i, B_i)$	Budget-aware loss ledger at stage i (definition 6.13)
Q_i	Confidence/quality profile of branch i (proposition 8.7)
$F(r, \tau)$	Fidelity of r for task τ (definition 10.2)
$\text{Rest}(\mathcal{R})$	Repair-stable domain (imported, <i>Observability Is Restoration</i>)
T	Task space; <i>open</i> if unbounded at production time
H	Enumerable hidden-state set (closed domains only, definition 10.1)
$M(c, \tau, t)$	Access-mode relation over consumer, task, time (proposition 10.2)
$D(A_i \Rightarrow A_j)$	Cumulative discard set between stages i and j (definition 10.3)
$C_i(d), I_i, \beta_i$	Stage-local repair cost, interference, and budget at relay stage i
$\{x_d, x_{\neg d}\}$	Minimal pair representing named distinction d (definition 10.4)
$\mathcal{C}_\tau(e)$	Correctness set of task τ evaluated against true object e
$r_{c,t}$	Substrate available to consumer c at time t (definition 10.5)
τ_d	Recovery task for distinction d (definition A.1)
T_{rec}	Witnessing family of recovery tasks (definition A.3)
$\theta(s)$	Resolution threshold for a map at scale 1: s (definition 10.6)

Appendix C

Relation to *Conservation of Ambiguity*: A Side-by-Side Comparison

Synopsis. This appendix compares the Composition Failure Theorem with the Redistribution Theorem of *Conservation of Ambiguity*, setting out their different assumptions, domains, and implications side by side, and arguing explicitly that neither is derivable from the other without the budget-relay machinery of Part II.

	Redistribution Theorem (theorem 4.6)	Composition (theorem 4.7)
Structure	Synchronic	Diachronic
Object	Single observer’s budget Φ	Sequence of budgets
Distinctions involved	Many, competing simultaneously	One (or none)
Conserved quantity	Total repair budget Φ_{tot}	None conserved
Coordination assumed	Full, by construction (one observer)	None, by construction
Failure mode	Local sharpening forces redistribution	Locally irreparable

C.1 Why One Does Not Reduce to the Other

definition 6.6's defining condition is that each stage's budget β_i is set without reference to any other stage's budget or ledger. It is worth checking directly what happens to Appendix A's witness if that condition is dropped and the stages' budgets are pooled into a single, simultaneously-allocated Φ_{tot} instead – not merely to confirm that theorem 4.6 then applies, but to see concretely what difference pooling makes to the actual outcome.

Recall the witness of construction A.5: five distinctions with costs 1, 2, 3, 2, 4, and two stage-local budgets $\beta_0 = 4$ and $\beta_1 = 3$, applied sequentially. Under definition 6.6, β_0 's unspent remainder (after funding $\{d_1, d_2\}$ at cost 3, one unit of β_0 goes unused, since no non-required distinction costs 1 or less) cannot be carried forward to supplement β_1 ; each stage's allocation is decided in isolation, and lemma A.4 shows d_5 is dropped at stage 0 regardless of construction.

Proposition C.1 (Pooling Can Restore Sufficiency). *Let $\Phi_{\text{tot}} = \beta_0 + \beta_1 = 7$ be made available to a single observer as one simultaneous allocation over all five distinctions of construction A.5, rather than as two separate stage-local budgets. Then a feasible allocation exists that funds $\{d_1, d_2, d_5\}$ in full, avoiding the insufficiency for τ_{d_5} that Appendix A shows is unavoidable under the separated relay.*

Proof. $C(d_1) + C(d_2) + C(d_5) = 1 + 2 + 4 = 7 = \Phi_{\text{tot}}$, so funding exactly $\{d_1, d_2, d_5\}$ is feasible under a single, simultaneous budget constraint of 7, using the full amount with nothing left over. Under this allocation, $d_5 \notin D(A_1)$, so by lemma A.2, sufficiency for τ_{d_5} is preserved. This allocation is unavailable to the separated relay of construction A.5 because $\beta_0 = 4$ alone is insufficient to fund $\{d_1, d_2, d_5\}$ (cost $7 > 4$), and proposition 6.12 forbids recovering d_5 once stage 0 has discarded it, regardless of how much budget β_1 later provides. $\square$

proposition C.1 is the concrete content behind the claim, made in prose in Chapter 9, that assuming a shared budget across all stages of a relay is precisely the condition under which theorem 9.2's construction becomes unavailable. It is stronger than merely showing the construction *fails to apply* under pooling: it exhibits an allocation that actively *succeeds* where the separated relay fails, using the identical total nominal resource. The insufficiency exhibited in Appendix A is therefore not a consequence of scarcity in any absolute sense – $\Phi_{\text{tot}} = 7$ was always, in total, enough to have funded $\{d_1, d_2, d_5\}$ – but a consequence specifically of definition 6.6's prohibition on transferring unspent capacity from one stage to a later one. This is the precise sense in which the two theorems occupy disjoint settings rather than one subsuming the other: theorem 4.6 presupposes exactly the transferability that theorem 9.2's witness depends on the absence of.

APPENDIX C. RELATION TO CONSERVATION OF
AMBIGUITY: A SIDE-BY-SIDE COMPARISON

Appendix D

Extended Worked Examples

Synopsis. This appendix sketches additional domains — textual transmission, oral tradition, and biological inheritance — that further test and illustrate the framework beyond the chapters given full treatment in Part IV. Treatment here is at appendix depth: each example is developed far enough to show the fit is real, not far enough to constitute a chapter.

D.1 Manuscript Transmission

A chain of scribal copies A_0 (the autograph) $\rightarrow A_1 \rightarrow \cdots \rightarrow A_n$ (an extant manuscript) is a substrate chain in the sense of definition 6.1, with each transition governed by a copyist's own budget: time, eyesight, familiarity with the source language, and attentiveness all bound β_i . Unlike Chapter 15's scale bar, a scribal error is essentially never declared: a miscopied word is presented exactly as confidently as a correctly copied one, making most manuscript corruption an instance of definition 2.8

rather than definition 2.7.

The chain structure is not merely formally convenient here; it is the literal basis of the scholarly discipline of textual criticism, which reconstructs a stemma — a family tree of copies — from surviving manuscripts and infers readings of the lost autograph by comparing independent branches.

Proposition D.1 (Branch Agreement as Evidence for the Archetype). *If two branches of a manuscript stemma, descended independently from a common ancestor, agree on a reading, and copying errors are independent across branches, that reading is more likely to reflect the ancestor than a reading found in only one branch.*

Proof Sketch. This is definition 8.2 applied to the manuscript domain: if errors in branch X and branch Y are independent, the probability that both branches independently introduce the *same* erroneous reading is far lower than the probability that both correctly preserve a shared true one. Agreement is therefore stronger evidence for the ancestor's reading than either branch's testimony alone. This is the formal content behind the standard editorial practice of preferring readings attested across independent branches of a stemma.

Where branches disagree, an editor constructing a critical text must choose a reading without being able to fully recover which branch, if either, preserved the ancestor correctly – a decision structurally similar to section 7.5's treatment of adjudication: the editor's judgment substitutes an operative reading for an underlying fact that may never be recoverable, though editorial judgment aims at recovering the truth in a way judicial adjudication does not always claim to.

D.2 Oral Tradition

Each retelling of an oral epic is a transition in the sense of proposition 19.2: a performer's schema, operating on whatever

residual trace of prior tellings they carry, produces a new performance. Oral-formulaic composition — fixed epithets, metrical formulae, repeated type-scenes, of the kind documented in the study of oral epic tradition — functions as a strong negative-interference device in the sense of definition 4.5: because performer and audience share the same formulaic expectations, maintaining a formula costs the performer far less than reconstructing an equivalent passage from scratch would, exactly as remark 14.1 showed for a painting's viewer.

This case surfaces a complication genuinely sharper than anything Part IV's chapters needed to confront. Chapter 6's root object (definition 6.8) presupposes some A_0 existing prior to the chain of transitions that follow it. An oral tradition without a fixed textual original may have no such A_0 at all: each performance is arguably a new production rather than a rendering of a prior fixed telling, in which case the entire preservation/reconstitution question of Part III may not have a privileged answer to converge toward, since there may be no single original for any performance's fidelity (definition 10.2) to be measured against. This monograph does not resolve this; it is flagged here as a further, distinct complication from the MEM|8 fork of Chapter 11, arising even under the narrow reading, since it concerns whether a root object exists at all rather than how faithfully a consumer accesses one.

D.3 Biological Inheritance

A caveat before this section. Conservation of Ambiguity reportedly applies repair-budget theory to biological evolution directly. The sketch below has not been checked against that treatment, and it is stated as an independent, appendix-level observation rather than as an extension building on it; a careful development of this case should verify the

relationship before treating either as complete.

Reproduction with recombination and mutation can be modeled as a substrate chain: a parental genome A_0 , an offspring genome A_1 produced by a transition governed by the fidelity of DNA repair and replication machinery — a budget in a sense closer to literal than metaphorical, since the relevant biological machinery is itself called repair machinery. Traits neutral under current selective pressure are exactly the non-required distinctions of definition 6.2: they are not funded preferentially, and under relaxed selection or genetic drift they can be lost across generations, exactly as proposition 6.12 describes cumulative discard accumulating along an uncoordinated chain.

Proposition D.2 (Environmental Change as Task Discovery). *If a trait is not required under the selective pressure τ_i operative during generations $0, \dots, n-1$, and a later environmental change constitutes a new selective task τ^* requiring that trait, the trait's loss under $\tau_0, \dots, \tau_{n-1}$ is unrecoverable by proposition 6.12, regardless of how strongly τ^* would now favor it.*

Proof Sketch. Immediate from treating successive generations as relay stages and environmental change as introducing an unanticipated task, exactly in the sense of proposition 3.7. This is the formal counterpart of a familiar evolutionary observation: a lineage cannot retroactively retain a trait it has already lost to drift or relaxed selection merely because a later environment would have favored it.

This proposition is offered cautiously, as a structural observation about the fit between the two frameworks rather than as a contribution to evolutionary theory, and its relationship to *Conservation of Ambiguity's* own treatment of the same domain remains, per the caveat above, to be checked.

Appendix E

Toward a Simulation Framework

Synopsis. This appendix outlines, without implementing, how budget relays could be represented computationally and investigated experimentally through simulation.

This appendix sketches, without implementing, how a budget relay (definition 6.6) could be simulated computationally. It is explicitly scoped as a direction for future computational work, not as part of the monograph’s formal content.

Each stage of a relay would be represented as a process holding a declared triple (C_i, I_i, β_i) , applying definition 6.2 to a set of input distinctions and producing an output rendering together with a loss ledger L_i per definition 6.13. A downstream query interface would then test theorem 9.2’s failure mode directly, by constructing task instances τ^* outside the set of tasks any stage was configured to anticipate, and checking whether the final rendering’s admissible continuation set $\mathcal{A}_{\tau^*}(A_n)$ contains the correct continuation.

Such a framework would allow empirical (rather than purely constructive) exploration of corollary 9.4’s conjectured growth

bound across relays of varying length, branching factor (chapter 8), and coordination mechanism (chapter 7), which is currently stated only as a conjecture in this monograph.

The monograph's own worked constructions supply natural first test cases rather than requiring new ones to be invented: construction A.5 and its extension (proposition A.8) give an exact, small instance against which a simulator's discard behavior could be checked directly, construction 20.1 gives a named, domain-flavored instance of the same size, and construction 8.4 gives the minimal fan-out case. A useful early milestone for any implementation would be reproducing these four constructions' discard sets exactly, before attempting the larger, randomized relays needed to explore corollary 9.4's growth-rate question empirically.

Appendix F

Open Questions Register

Synopsis. This appendix maintains a single consolidated record of every conjecture and open question raised in the body, keeping a clear separation between established results and speculative extensions.

Per the epistemic-labeling discipline stated in the front matter, this appendix consolidates every claim in the body labeled Conjecture or Open Question, together with every substantive open item flagged informally in a Remark rather than housed in a formal environment, kept separate from proved content so that the monograph's unproved commitments are never left implicit. Entries are grouped by the part of the framework they bear on, rather than by chapter order, since several recur across chapters far apart in the text.

- **The MEM|8 fork** (open Question 11.1, Ch. 11). The single largest open dependency of the monograph: whether MEM|8's claim generalizes from biological recall specifically (narrow reading) to representation as such (broad reading), and, if the latter, whether Part I requires revision from the ground up. open Question 10.8 (Ch. 10) is

the same fork applied to the specific question of whether any instance of “direct access” involves strictly zero mediation.

- **Discard-rule robustness — partially resolved** (Ch. 6, remark 6.5; Appendix A). definition 6.2 is satisfied by more than one non-equivalent construction — at minimum construction 6.3 and construction 6.4 — and the two can select different discard sets for the same budget and priorities in general. lemma A.4 resolves this specifically for theorem 9.2’s own proof, by exhibiting a witness relay whose key discard is forced under any construction satisfying definition 6.2, but only for stages where the required distinctions’ cost closely exhausts the local budget (**non-exhausting relays**, below). What remains open is whether every *other* claim in this monograph stated to hold for “every” budget relay enjoys the same construction-independence, or whether some depend, unexamined, on a specific choice of local discard rule.
- **Non-exhausting relays** (Appendix A). lemma A.4 guarantees construction-independent discard only for stages where the required distinctions’ cost closely exhausts the local budget. Whether theorem 9.2’s conclusion is robust to choice of construction for relays *without* this near-exhaustion property — where genuine slack remains after funding every required distinction — is open; construction 6.3 and construction 6.4 may disagree on which non-required distinctions such slack goes to, with unexamined consequences for which future tasks such a relay turns out to support.
- **Root-preservation bootstrapping** (Ch. 6, remark 6.10). Preserving a root object (definition 6.8) so that proposition 6.9 remains available is itself a repair decision that must be funded before the future task space justifying

it is known. Whether this can be given a more precise decision-theoretic treatment — for instance, as an expected-value calculation over a distribution of possible future tasks — rather than being left at the level of a qualitative design recommendation, is open.

- **Infinite relays** (Ch. 6, definition 6.1). This monograph treats only substrate chains with finitely many stages. Whether an infinite chain behaves differently in kind, rather than merely in degree, from the finite case — for instance, whether proposition 6.12 or theorem 9.2 continue to hold, require additional hypotheses, or fail outright in the infinite setting — is unaddressed.
- **Branching relays** (Appendix A, Ch. 8). Appendix A's construction is linear (definition 6.1) throughout. Whether an analogous construction-independent witness exists for the branching relays of Chapter 8, given the additional complication of proposition 8.6, has not been checked.
- **Quantitative tightness of corollary 9.4** (Ch. 9, Appendix A). Appendix A establishes only the existence half of this corollary — that cumulative discard is non-decreasing along the witnessing construction, realized with equality across arbitrarily many pass-through stages — and explicitly does not establish the stronger claim that cumulative discard *risk* grows across relays in general, nor whether a sharper bound holds under specific coordination mechanisms.
- **General coordination protocols** (Ch. 7, Ch. 21). Whether a coordination protocol could be designed to bound, though not eliminate, theorem 9.2's failure mode across an entire class of relays, rather than only via the domain-specific mechanisms of Chapter 7. proposition C.1 (Appendix C) shows that one particular, more radical move

- pooling stage budgets rather than coordinating separate ones, per Section 7.7's distinction — can restore sufficiency in a specific witness; it does not answer this broader question, since pooling eliminates the stage-separation the question is asking about coordinating *within*, rather than supplying a coordination protocol in the sense of Chapter 7's taxonomy.
- **The closing synthesis** (conjecture 21.1, Ch. 21). Labeled a conjecture because the unification of theorem 9.2 with the fidelity continuum has not itself been given an independent proof beyond what its components separately establish.

Bibliography Note

Placeholder: to be developed if an external reference list becomes necessary beyond the inline citations used throughout the body.